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Cosmic Radiation

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Abstract

In this experiment, we explore the nature of cosmic radiation, with a particular focus on muons, which represent the dominant component of cosmic rays at sea level. The primary goal is to detect these highly energetic particles and investigate their key physical properties. The study is carried out in two stages. The first stage involves familiarizing with the experimental setup and performing a careful calibration of the electronics through combined measurements of time and energy. This calibration lays the foundation for reliable data acquisition. In the second stage, the calibrated system is used to record various spectra, enabling the determination of muon energy loss in matter as well as the angular dependence of their flux. Through these measurements, we gain valuable insights into the interaction mechanisms of cosmic muons and the factors influencing their detection. The experiment also serves as a practical introduction to techniques used in experimental particle physics.

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1 Introduction

Cosmic radiation consists of high-energy particles that travel through space and eventually reach Earth. These particles originate from various astrophysical sources such as the Sun, supernovae, and distant galaxies. When these primary cosmic rays enter Earth's atmosphere, they collide with atmospheric nuclei, initiating a chain of reactions known as extensive air showers. Through these interactions, a large number of secondary particles are created.

The air showers include different types of particles: photons, electrons, positrons, hadrons, and muons. Muons, in particular, play a central role because of their ability to penetrate the atmosphere and reach the ground due to their relatively long lifetime and high energy. At sea level, muons make up the majority of cosmic radiation, accounting for nearly 90% of the particles detected.

Studying muons not only helps us understand the processes happening in the upper atmosphere but also offers a useful way to explore the interaction of high-energy particles with matter. In this experiment, the goal is to detect cosmic muons and investigate their key properties, such as how they lose energy when passing through matter, how accurately they can be timed and located, and how their detection rate depends on the angle at which they enter the detector.

To achieve these measurements, we use a setup that consists of plastic scintillator plates coupled with photomultiplier tubes (PMTs). When a muon passes through the scintillator, it loses energy and produces a flash of light. This light is collected by the PMTs and converted into electrical signals. The signals are then processed using amplifiers, discriminators, coincidence circuits, and data acquisition systems, allowing us to measure both the deposited energy and the time information.

The experiment is divided into two main parts. First, the detector system is carefully calibrated to ensure accurate measurements. This involves studying the PMT signals, performing time and energy calibrations, and checking for possible effects from background light. Once the calibration is complete, we carry out several measurements that include determining the energy loss of muons in the scintillator, analyzing how the light signal attenuates within the scintillator material, measuring the effective speed of light inside the scintillator, evaluating the timing and spatial resolution of the detector, and finally, studying how the intensity of muons varies with different angles of incidence.

2 Theoretical Background

Before performing the experiment, it is essential to understand key concepts related to Cosmic Radiation.

2.1 Cosmic Radiation: Origin, Composition, and Air Showers

Cosmic radiation refers to high-energy particles that continuously arrive at Earth from space. These particles are generated in various astrophysical processes such as solar activity, supernova explosions, active galactic nuclei, and possibly even more exotic sources in the universe [1, 19, 7]. On their way to Earth, these particles travel vast distances, often interacting with interstellar magnetic fields, but many of them still manage to reach our atmosphere.

When discussing cosmic radiation, a distinction must be made between *primary* and *secondary* cosmic radiation.

2.1.1 Primary Cosmic Radiation

Primary cosmic rays are those particles that reach the Earth's atmosphere directly from outer space. They are composed predominantly of protons (about 79%), followed by helium nuclei (around 15%) and trace amounts of heavier nuclei such as carbon, oxygen, nitrogen, and iron [1, 11, 13]. These particles possess extremely high energies, ranging from a few GeV up to energies exceeding 10^{20} eV [19].

2.1.2 Secondary Cosmic Radiation

When primary cosmic rays enter the Earth's atmosphere, they interact with atmospheric nuclei, initiating a cascade of nuclear reactions. These collisions produce a wide range of secondary particles, including pions, kaons, neutrons, protons, and heavier mesons. Many of these secondary particles are unstable and quickly decay into more stable particles such as muons, electrons, positrons, neutrinos, and gamma rays [9, 19, 7]. This complex cascade of particle interactions and decays is known as an *extensive air shower* [1].

By the time the shower reaches sea level, most of the unstable particles have already decayed, and the dominant component detected at the surface consists of muons, which account for nearly 90% of the particles detected [11, 13].

2.2 Development of Cosmic Air Showers

The secondary particle cascades in the atmosphere can be divided into two major components: electromagnetic and hadronic showers.

2.2.1 Electromagnetic Showers

Electromagnetic showers are initiated by high-energy photons (γ) or electrons ($e^\pm$) entering the Earth's atmosphere. These showers develop primarily through two fundamental physical processes: pair production and bremsstrahlung [1, 11].

Pair Production When a high-energy photon passes close to an atomic nucleus, its electromagnetic field can cause the photon to convert into an electron-positron pair:

$$\gamma + Z \rightarrow e^- + e^+ + Z.$$

This process requires the photon energy to exceed twice the electron rest mass energy, i.e., $E_\gamma > 2m_e c^2 \approx 1.022$ MeV, where m_e is the electron mass and c is the speed of light. The nucleus serves as a recoil partner to conserve momentum during this interaction.

Bremsstrahlung The newly created electrons and positrons, being charged, are deflected by the Coulomb fields of atmospheric nuclei, emitting further photons via bremsstrahlung radiation:

$$e^\pm + Z \rightarrow e^\pm + \gamma + Z.$$

The energy loss due to bremsstrahlung becomes increasingly significant at higher electron energies, and the probability of photon emission grows roughly proportionally to the electron energy.

Cascade Development These two processes alternate repeatedly: photons generate electron-positron pairs, and these charged particles emit further bremsstrahlung photons, which again produce new pairs. This repeating sequence leads to an exponential growth of particles in the shower, forming a cascade.

The number of particles continues to grow until the average particle energy drops below a certain critical energy E_c . Below this threshold, ionization losses dominate, and further particle multiplication essentially stops. The critical energy for air is approximately $E_c \approx 85$ MeV [11].

Shower Maximum and Longitudinal Development The longitudinal development of the shower — how the particle number evolves with atmospheric depth — can be approximated by the gamma distribution:

$$\frac{dE}{dt} = E_0 \cdot b \cdot \frac{(bt)^{a-1}}{\Gamma(a)} e^{-bt},$$

where:

- E_0 is the initial energy of the primary photon or electron,
- $t = X/X_0$ is the atmospheric depth in units of radiation length X_0 ,
- a and b are shape parameters fitted from experimental data,
- $\Gamma(a)$ is the gamma function [1].

The point at which the maximum number of particles is reached — called the shower maximum — depends logarithmically on the initial energy:

$$t_{\max} \approx \ln \left(\frac{E_0}{E_c} \right).$$

Lateral Spread In addition to longitudinal development, the shower also spreads laterally due to multiple Coulomb scattering of electrons and positrons. However, most of the particles remain concentrated near the shower axis.

End of Shower When particle energies fall below the critical energy E_c , ionization losses dominate:

$$\left(\frac{dE}{dx} \right)_{\text{ion}} \propto \frac{1}{\beta^2} \ln \left(\frac{2m_e c^2 \beta^2 \gamma^2}{I} \right),$$

where I is the mean excitation potential of the medium, and β and γ are the usual relativistic factors [11, 13].

At this stage, the number of particles rapidly diminishes, and the shower essentially terminates.

2.2.2 Hadronic Showers

Hadronic showers are initiated when a high-energy primary cosmic ray, typically a proton or a heavier nucleus, collides with an atomic nucleus in the Earth's atmosphere. These collisions produce a large number of secondary hadrons, primarily pions ($\pi^\pm$, π^0), kaons ($K^\pm$, K^0), nucleons, and other mesons and baryons [9, 13, 19, 1].

Primary Interaction The initial collision between the primary cosmic ray and the atmospheric nucleus is governed by the strong interaction, resulting in the production of multiple secondary particles in a highly inelastic process:

$$p + N \rightarrow \pi^\pm, \pi^0, K^\pm, K^0, N, \bar{N}, \dots$$

where N denotes nucleons (protons and neutrons).

Hadronic Cascade The secondary hadrons produced can interact again with other nuclei in the atmosphere, creating additional hadronic interactions and producing even more particles. This leads to a complex hadronic cascade where both the number and energy of secondary particles evolve with each generation.

The multiplicity of produced pions in these interactions roughly scales with the energy of the incident particle, and after several generations, most of the energy of the original primary particle is distributed among a large number of lower-energy secondaries.

Pion Decay: Connection to Muons The pions created in the hadronic cascade play a central role in the production of muons:

- **Neutral pions** (π^0) decay very quickly into two photons, feeding the electromagnetic component of the air shower:

$$\pi^0 \rightarrow \gamma + \gamma.$$

These high-energy photons may then initiate electromagnetic showers through pair production and bremsstrahlung.

- **Charged pions** ($\pi^\pm$) can either interact again or decay into muons and muon neutrinos:

$$\pi^+ \rightarrow \mu^+ + \nu_\mu, \quad \pi^- \rightarrow \mu^- + \bar{\nu}_\mu.$$

The competition between decay and further interaction depends on the atmospheric density and the pion energy. The probability for decay increases as the atmosphere becomes thinner at higher altitudes or as the pion energy decreases.

Kaon Decay Kaons, though produced less abundantly than pions, contribute additional muons via similar decay channels:

$$K^+ \rightarrow \mu^+ + \nu_\mu, \quad K^- \rightarrow \mu^- + \bar{\nu}_\mu.$$

However, due to their shorter lifetime and lower production rate, kaons play a smaller role compared to pions in the overall muon flux [11, 1].

Longitudinal Development The longitudinal profile of hadronic showers is more irregular compared to electromagnetic showers due to the stochastic nature of inelastic collisions and particle production. Nonetheless, one can describe the shower development approximately using the concept of interaction length λ_I , which represents the mean distance between two successive hadronic interactions:

$$\lambda_I = \frac{A}{N_A \cdot \sigma_{inel}},$$

where:

- A is the molar mass of air,
- N_A is Avogadro's number,
- σ_{inel} is the inelastic cross section of the interaction.

Typically, $\lambda_I \approx 80 - 120 \text{ g/cm}^2$ for air depending on the incident energy [13, 11].

Energy Dissipation With each interaction, a fraction of the energy is transferred to the electromagnetic component through π^0 decay, while the remainder continues to feed the hadronic cascade and muon production. As a rule of thumb, approximately one-third of the energy is passed to the electromagnetic component at each generation.

Muon Propagation The produced muons are highly penetrating due to their small cross-section for interaction and their relativistic velocities. Their relatively long lifetime of $\tau_\mu \approx 2.2 \mu\text{s}$ allows many muons to reach the Earth's surface before decaying:

$$\mu^\pm \rightarrow e^\pm + \nu_e(\bar{\nu}_e) + \bar{\nu}_\mu(\nu_\mu).$$

This is why muons dominate the particle flux observed at sea level, making them ideal for ground-based cosmic ray detection experiments [11, 13, 7].

Summary of Hadronic Showers To summarize, hadronic showers involve:

- Primary high-energy nucleon collisions producing secondary hadrons.
- Multiple generations of hadronic interactions creating complex particle cascades.
- Decays of pions and kaons producing muons and neutrinos.
- Transfer of energy from hadronic to electromagnetic components via π^0 decays.

Hadronic and electromagnetic components are strongly coupled, and together they shape the full development of cosmic air showers.

2.3 Muon Production and Energy Loss in Scintillator

2.3.1 Production of Muons in Cosmic Ray Air Showers

Muons are produced as part of the secondary cosmic radiation through a sequence of high-energy interactions that occur when primary cosmic rays collide with atomic nuclei in the Earth's atmosphere [19, 1, 13].

The process begins when a primary cosmic ray, typically a high-energy proton or a heavy nucleus, strikes a nucleus in the upper atmosphere. This interaction leads to a hadronic cascade, where many secondary particles are generated, including pions ($\pi^\pm, \pi^0$) and kaons ($K^\pm, K^0$). The neutral pions decay very rapidly into high-energy photons, which contribute to the electromagnetic shower. However, the charged pions are responsible for the majority of muon production.

The dominant decay channel for charged pions is:

$$\pi^+ \rightarrow \mu^+ + \nu_\mu, \quad \pi^- \rightarrow \mu^- + \bar{\nu}_\mu.$$

Kaons contribute similarly through:

$$K^+ \rightarrow \mu^+ + \nu_\mu, \quad K^- \rightarrow \mu^- + \bar{\nu}_\mu.$$

Since these decays involve only weak interactions, the pions and kaons have sufficient time to decay into muons before undergoing further interactions, especially at higher altitudes where the atmosphere is thinner. The produced muons are highly penetrating due to their long lifetime ($\tau_\mu \approx 2.2 \mu\text{s}$) and relativistic speeds, allowing them to reach ground level where they can be detected experimentally [11, 7, 13].

At sea level, approximately 90% of the cosmic rays detected consist of muons, making them the dominant component of secondary cosmic radiation [1, 19].

2.3.2 Energy Loss of Muons in Scintillator

As muons pass through matter, including detector materials such as plastic scintillators, they primarily lose energy through ionization and excitation of the atoms in the medium. This energy loss can be accurately described using the Bethe-Bloch equation [11, 9, 13].

The general form of the Bethe-Bloch formula is:

$$-\frac{dE}{dx} = \frac{4\pi N_A r_e^2 m_e c^2}{\beta^2} \cdot \frac{Z}{A} \cdot \left[\frac{1}{2} \ln \left(\frac{2m_e c^2 \beta^2 \gamma^2 T_{\max}}{I^2} \right) - \beta^2 - \frac{\delta(\beta\gamma)}{2} \right]$$

where:

- N_A is Avogadro's number,
- r_e is the classical electron radius,
- m_e is the electron mass,
- c is the speed of light,
- Z and A are the atomic number and atomic mass of the medium,
- $\beta = v/c$ is the relative velocity,
- $\gamma = 1/\sqrt{1 - \beta^2}$ is the Lorentz factor,
- $T_{\max}$ is the maximum energy transfer in a single collision,
- I is the mean excitation potential,
- $\delta(\beta\gamma)$ is the density effect correction.

For typical relativistic cosmic muons (kinetic energy of a few hundred MeV), the stopping power simplifies to approximately the minimum ionizing value.

Scintillator Composition and Parameters

The scintillator used in this experiment is plastic scintillator, which can be approximated by polystyrene (C_8H_8), consisting of hydrogen and carbon. The relevant material parameters are [11, 16]:

- Density: $\rho = 1.032 \text{ g/cm}^3$,
- Effective $Z/A \approx 0.541$,
- Mean excitation potential: $I \approx 68 \text{ eV}$.

Energy Loss Calculation

Using typical muon energies of $E_\mu \approx 300 \text{ MeV}$ (consistent with cosmic ray muons at sea level [1]), and applying the Bethe-Bloch equation for this energy range, one finds that the energy loss is approximately:

$$-\frac{dE}{dx} \approx 2 \text{ MeV} \cdot \text{g}^{-1} \cdot \text{cm}^2.$$

Multiplying by the density of the scintillator gives:

$$-\frac{dE}{dx} \approx 2 \text{ MeV} \cdot \text{g}^{-1} \cdot \text{cm}^2 \times 1.032 \text{ g/cm}^3 = 2.064 \text{ MeV/cm}.$$

For a scintillator thickness of $x = 2 \text{ cm}$, the total energy loss is:

$$\Delta E = \left(\frac{dE}{dx} \right) \cdot x = 2.064 \text{ MeV/cm} \times 2 \text{ cm} = 4.128 \text{ MeV}.$$

This result is consistent with theoretical expectations.

Energy Loss Curve

The energy loss of muons in matter is strongly dependent on their velocity and kinetic energy. As described by the Bethe-Bloch equation, the stopping power $-dE/dx$ decreases with increasing energy at low energies, reaches a minimum, and then slowly rises again due to relativistic effects at very high energies [11, 13].

At low kinetic energies (below a few tens of MeV), muons lose energy primarily through ionization with relatively high stopping power. As the muon energy increases, the ionization loss decreases

due to the $1/\beta^2$ dependence of the Bethe-Bloch formula, where $\beta = v/c$. This leads to a characteristic broad minimum in the stopping power, typically occurring in the energy range of about 200 to 300 MeV for most materials, including plastic scintillators.

In this minimum region, muons are referred to as *minimum ionizing particles* (MIPs), where $-dE/dx$ reaches its lowest value, making this energy region particularly relevant for cosmic ray muons at sea level. This is because the typical mean energy of atmospheric muons reaching the surface lies approximately in the MIP region [1].

For higher energies (above a few GeV), the energy loss starts to rise again slowly due to relativistic effects such as density corrections and radiative losses (bremsstrahlung, pair production, and photonuclear interactions). However, these radiative processes remain negligible for muon energies below several hundred GeV in typical detector materials [11].

A schematic illustration of this behavior would show:

- A sharp decrease in $-dE/dx$ at low energies.
- A wide, flat minimum around 200–300 MeV.
- A slow logarithmic rise at high energies due to relativistic corrections.

For the plastic scintillator used in this experiment, the minimum ionizing energy loss is approximately $2 \text{ MeV g}^{-1} \text{ cm}^2$, which translates to about 2.06 MeV/cm for its density of $\rho = 1.032 \text{ g/cm}^3$.

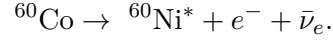
This nearly constant energy loss in the MIP region is one of the reasons why plastic scintillators are highly effective for muon detection: the detector response remains stable for a wide range of muon energies commonly observed in cosmic rays.

2.4 Radioactive Decays: ^{60}Co and ^{22}Na

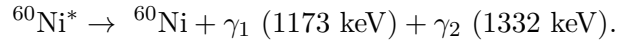
In this experiment, two radioactive sources are used for calibration and energy determination: Cobalt-60 (^{60}Co) and Sodium-22 (^{22}Na). Both isotopes emit γ -radiation, which interacts with the scintillator and allows calibration of the detector via Compton scattering processes [11, 9, 13, 19].

Decay of ^{60}Co

Cobalt-60 is a β^- emitter with a half-life of approximately 5.27 years. The decay proceeds as:



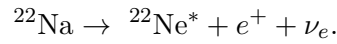
The daughter nucleus $^{60}\text{Ni}^*$ is produced in an excited state and de-excites via emission of two successive γ -rays:



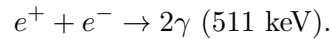
Both γ -lines are very close in energy and are emitted almost simultaneously. In the context of this experiment, these two energies are both relevant for energy calibration because both contribute to the observed Compton spectra.

Decay of ^{22}Na

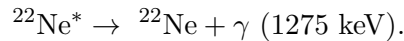
Sodium-22 undergoes β^+ decay and electron capture, leading to the production of excited Neon-22 nuclei:



The emitted positron rapidly annihilates with an electron, producing two annihilation photons of energy:



In addition, the excited Neon-22 de-excites by emitting a further γ -ray:



For this experiment, the 511 keV annihilation photons are mainly used for energy calibration purposes because they form a well-defined peak in the energy spectrum.

Compton Edge Calculation

When γ -photons interact with the scintillator material, one of the dominant processes at these photon energies is Compton scattering. In such interactions, the incident photon transfers a portion of its energy to an atomic electron, causing the electron to recoil while the photon is scattered with reduced energy. The amount of energy transferred depends on the scattering angle θ between the incident and scattered photon directions [11, 9, 13].

The energy of the scattered photon E'_γ can be calculated using the Compton scattering formula:

$$E'_\gamma = \frac{E_\gamma}{1 + \frac{E_\gamma}{m_e c^2}(1 - \cos \theta)},$$

where E_γ is the energy of the incoming photon, and $m_e c^2 = 511$ keV is the rest mass energy of the electron.

Since energy is conserved, the energy transferred to the recoiling electron (i.e., the kinetic energy E_e of the electron) is given by:

$$E_e = E_\gamma - E'_\gamma.$$

The maximum possible energy transfer — which defines the *Compton edge* — occurs when the photon is backscattered, i.e., when $\theta = 180^\circ$. Substituting this into the formula, the expression for the maximum electron energy becomes:

$$E_{e,\max} = E_\gamma \cdot \left(1 - \frac{1}{1 + \frac{2E_\gamma}{m_e c^2}} \right).$$

This Compton edge represents the highest energy that can be deposited in the scintillator by a single Compton event for a given photon energy E_γ . In practice, the position of the Compton edge is clearly visible in the measured energy spectra and serves as an important reference point for energy calibration.

We now calculate the Compton edges for both sources used in this experiment.

Compton Edge for ^{60}Co The ^{60}Co source emits two prominent γ -lines at 1173 keV and 1332 keV. For simplicity, we approximate the Compton edge using their mean energy:

$$E_{\gamma,\text{mean}} = \frac{1173 + 1332}{2} = 1252.5 \text{ keV}.$$

Inserting into the Compton edge formula:

$$\begin{aligned} E_{e,\text{max}} &= 1252.5 \text{ keV} \cdot \left(1 - \frac{1}{1 + \frac{2 \cdot 1252.5}{511}} \right) \\ &= 1252.5 \text{ keV} \cdot \left(1 - \frac{1}{1 + 4.902} \right) \\ &= 1252.5 \text{ keV} \cdot \left(1 - \frac{1}{5.902} \right) \\ &= 1252.5 \text{ keV} \cdot 0.8306 \\ &\approx 1040.7 \text{ keV}. \end{aligned}$$

Thus, the Compton edge for ^{60}Co is approximately 1041 keV.

Compton Edge for ^{22}Na For the ^{22}Na source, we use the 511 keV annihilation photons:

$$\begin{aligned} E_{e,\text{max}} &= 511 \text{ keV} \cdot \left(1 - \frac{1}{1 + \frac{2 \cdot 511}{511}} \right) \\ &= 511 \text{ keV} \cdot \left(1 - \frac{1}{1 + 2} \right) \\ &= 511 \text{ keV} \cdot \left(1 - \frac{1}{3} \right) \\ &= 511 \text{ keV} \cdot \frac{2}{3} \\ &\approx 340.7 \text{ keV}. \end{aligned}$$

Thus, the Compton edge for ^{22}Na is approximately 341 keV.

Isotope	Compton Edge (keV)
^{60}Co	1041
^{22}Na	341

These calculated Compton edges are later used as reference points for the linear energy calibration of the scintillator system during the experiment [11, 9, 13].

2.4.1 Landau Distribution and Its Importance in This Experiment

When a charged particle such as a muon traverses matter, it interacts with atomic electrons and loses energy mainly via ionization and excitation. The average energy loss per unit path length can be described by the Bethe-Bloch formula. However, the actual energy loss in each event fluctuates around this mean due to the stochastic nature of the microscopic collision processes [11, 1, 13].

In thick absorbers, these fluctuations tend to average out due to the central limit theorem, resulting in approximately Gaussian-shaped energy loss distributions. However, in thin absorbers — like the 2 cm plastic scintillator used in this experiment — only a small number of collisions contribute significantly to the total energy loss. This makes rare large-energy-transfer collisions more prominent, leading to highly asymmetric distributions.

These rare events occur when a muon transfers a large amount of energy to a single electron, producing a so-called *delta ray*, i.e., an energetic secondary electron capable of traveling a considerable distance on its own.

The statistical distribution describing these energy loss fluctuations in thin materials was first derived by Lev Landau in 1944. The Landau distribution has two key characteristics:

- A sharp peak at the *most probable value* (MPV), representing the most likely energy loss.
- A long high-energy tail extending toward larger energy losses, caused by the occasional production of delta-rays.

The probability density function (PDF) of the Landau distribution is defined as:

$$f(\lambda) = \frac{1}{\pi} \int_0^\infty \exp(-t \ln t - \lambda t) \sin(\pi t) dt,$$

where λ is a dimensionless, normalized variable related to the energy loss.

In practical data analysis, a simplified approximation known as the *Moyal distribution* is often used to fit experimental energy spectra because it closely resembles the Landau shape but is easier to handle numerically [11, 7].

For this experiment, the Landau distribution plays a crucial role for two primary reasons:

1. **Energy Calibration:** The ADC spectra obtained from muons passing through the scintillator exhibit Landau-shaped distributions. Fitting these spectra with a Landau or Moyal function allows for accurate determination of the most probable energy loss, which is used as a reference point for energy calibration.
2. **Thin Detector Response:** Since our scintillator thickness is relatively small, the number of collisions is limited. As a result, the fluctuation effects described by the Landau distribution are pronounced and must be properly accounted for when evaluating energy loss and detector performance.

2.4.2 Light-Matter Interaction and Attenuation Law

When electromagnetic radiation passes through matter, it interacts with the material in various ways depending on the photon energy, the atomic structure of the medium, and the thickness of the absorber. These interactions can result in absorption or scattering processes, leading to a reduction in the intensity of the transmitted beam [11, 13, 9, 1, 7].

At photon energies relevant to this experiment (i.e., γ -rays from ^{60}Co and ^{22}Na), three primary interaction mechanisms dominate:

1. **Photoelectric Effect:** The photon is fully absorbed, and its energy is transferred to an atomic electron, which is ejected from the atom. This process is dominant at lower photon energies and exhibits a strong dependence on the atomic number Z of the absorber material [11].
2. **Compton Scattering:** The photon transfers part of its energy to an electron via elastic scattering, while the photon itself continues with reduced energy and altered direction. This process dominates at intermediate photon energies, which is the relevant energy regime for this experiment [13, 9].
3. **Pair Production:** At higher photon energies ($E_\gamma > 1.022 \text{ MeV}$), photons can transform into an electron-positron pair in the electric field of a nucleus. This process becomes significant for higher energy photons but plays only a minor role in our experiment due to the energy range of the sources used [1, 7].

Regardless of the specific interaction type, the combined effect of these processes leads to a gradual reduction in photon beam intensity as the radiation passes through matter. This reduction follows the empirical **attenuation law**, often called the **Beer-Lambert law**, which is mathematically expressed as:

$$I(x) = I_0 \cdot e^{-\mu x},$$

where:

- $I(x)$ is the transmitted photon intensity after passing through a material thickness x ,
- I_0 is the initial photon intensity,
- μ is the linear attenuation coefficient (in units of cm^{-1}), which depends on both the material properties and the photon energy [11, 16].

The total attenuation coefficient μ can be written as the sum of the partial contributions from the three interaction mechanisms:

$$\mu = \mu_{\text{photo}} + \mu_{\text{Compton}} + \mu_{\text{pair}}.$$

In plastic scintillators such as polystyrene, and at the γ -ray energies used in this experiment, Compton scattering dominates, making μ_{Compton} the primary contributor to μ [9, 13, 11].

The attenuation law plays an important role in several parts of this experiment:

- **Scintillation Light Propagation:** After muons deposit energy in the scintillator, the generated light signal propagates through the material and is partially attenuated before reaching the photomultiplier tubes (PMTs). Understanding this attenuation is essential for correct interpretation of the light yield and for determining the attenuation length of the scintillator material.
- **Calibration Sources:** The Beer-Lambert law also describes how the intensity of γ -photons from calibration sources diminishes as they traverse the scintillator, contributing to the observed energy spectra.

By analyzing the attenuation behavior, we can extract valuable information about the optical properties of the scintillator and ensure accurate calibration and signal interpretation throughout the experiment.

2.5 Main Detector and Electronic Components

The hardware employed in this experiment forms a coherent and well-synchronized detection chain, beginning with particle interaction and ending in digital event recording. Figure 2.1 (adapted from [6]) illustrates the complete arrangement of detectors and electronics.

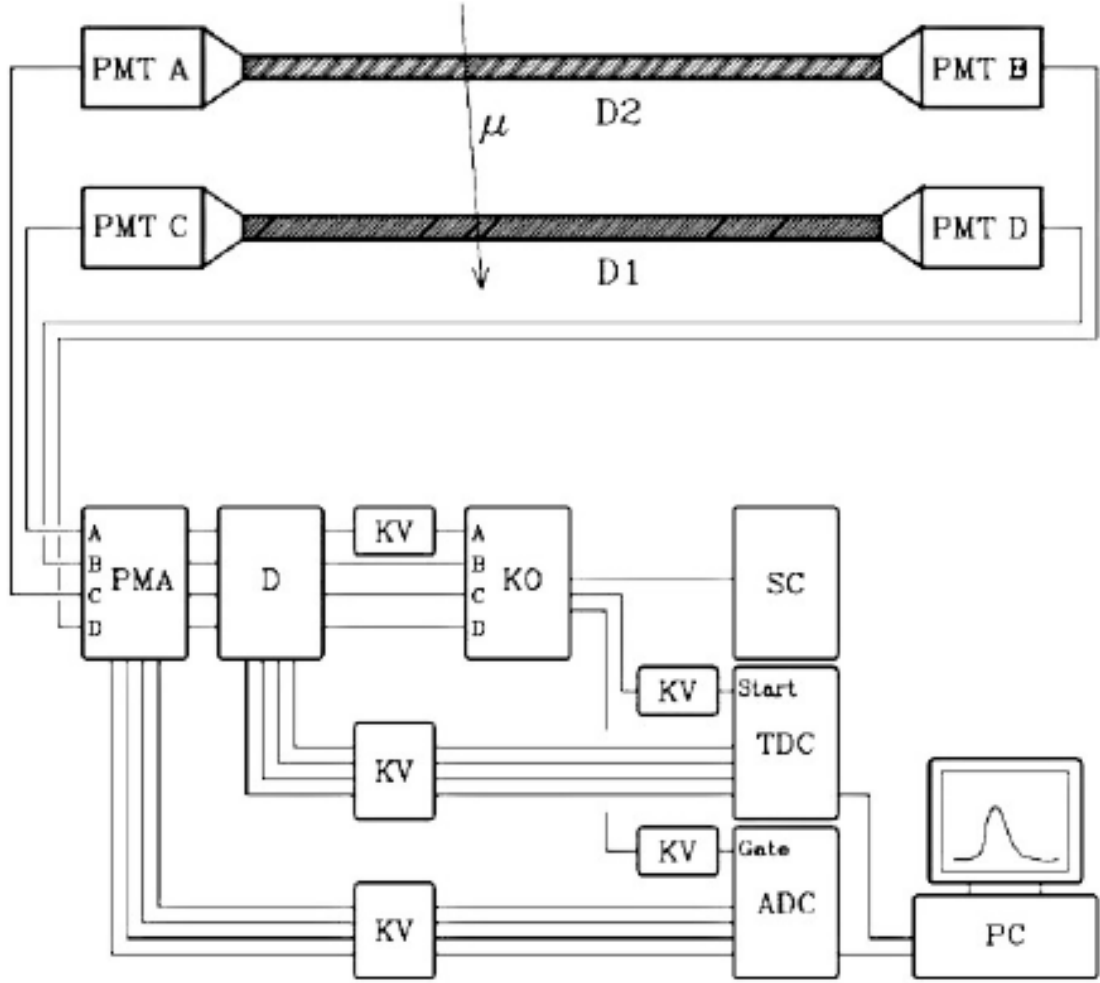


Figure 2.1: Circuit of the measurement setup. The system includes plastic scintillators (D1, D2), light guides, PMTs, amplifiers, discriminators, a coincidence unit, ADC, TDC, scaler, and delay modules. Adapted from [6].

2.5.1 Scintillator

Plastic scintillators are fast (decay time 2 ns) organic compounds that convert ionizing particle energy into visible photons. When charged particles pass through, they excite aromatic molecules; the de-excitation yields an average of 1 photon per 100 eV deposited [3]. Light output scales linearly with deposited energy:

$$N = \xi \cdot \frac{E}{h\nu}$$

where typically $\xi \approx 0.01$ for plastic scintillators [8]. Figure 2.2 shows the scintillation and light transport process.

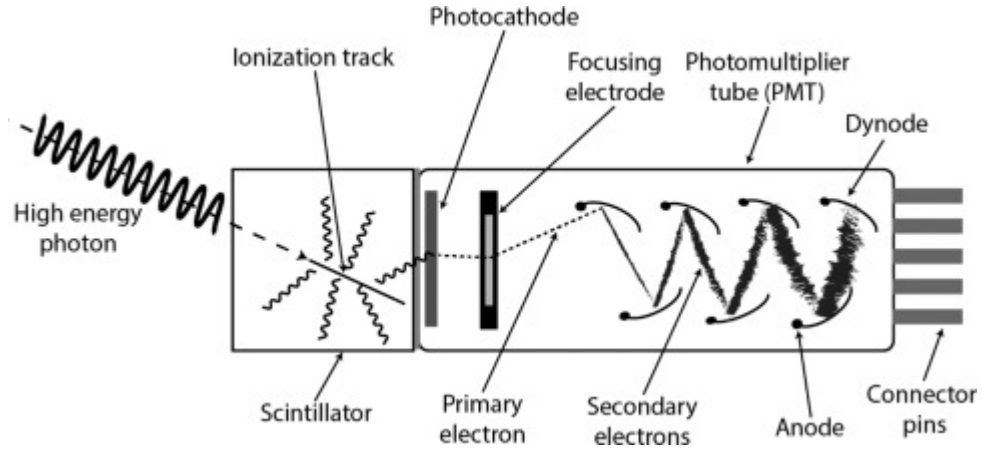


Figure 2.2: Schematic illustration of the scintillation process inside plastic scintillators. High-energy charged particles ionize molecules, leading to photon emission via de-excitation. These photons are then transported through the light guide toward the PMT. Adapted from [4, 8].

2.5.2 Light Guide

Light guides, commonly polished acrylic or polystyrene, funnel photons from the scintillator into the PMT window with minimal loss (total internal reflection). Proper optical coupling is crucial; losses can reduce detection efficiency by 10–30% if neglected [5].

2.5.3 Photomultiplier Tube (PMT)

PMTs convert a few photons into detectable electrical pulses via a series of dynodes. A photon releases a photoelectron at the photocathode; this electron multiplies by factors of 4–5 per dynode stage, leading to overall gains of 10^6 – 10^7 [8, 14]. Figure 2.3 depicts the internal structure.

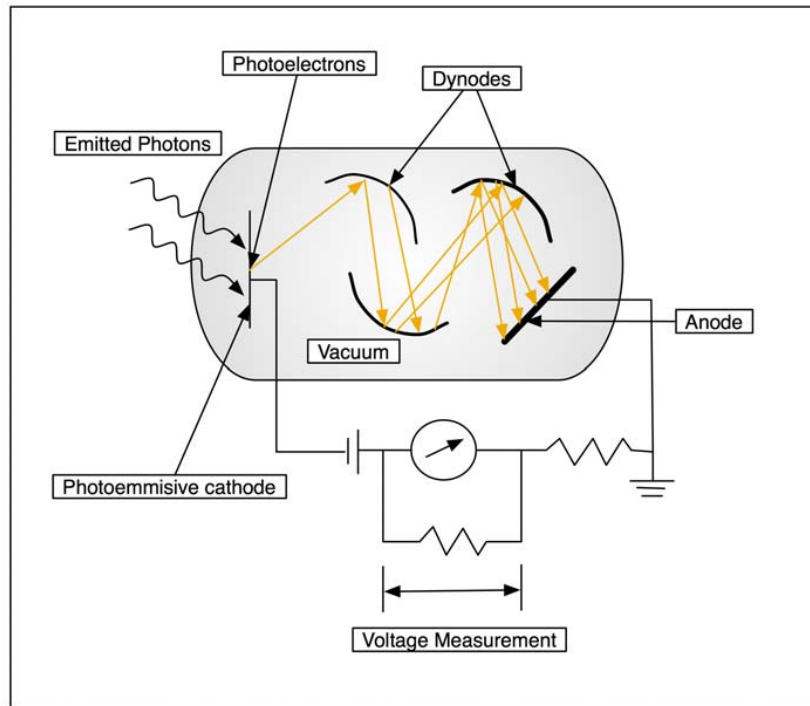


Figure 2.3: Schematic diagram of the working principle of a photomultiplier tube (PMT). Incident photons release photoelectrons from the photocathode, which are then multiplied through a chain of dynodes and collected at the anode, producing an amplified current signal. Adapted from [15, 8].

2.5.4 Amplifier (PMA)

The PMA chain first pre-amplifies ($\times 10$ – $\times 100$) the PMT signal and then shapes it (30 ns shaping time [18]). This prepares the signal for discrimination and digitization.

2.5.5 Discriminator (D)

Analog pulses are fed into discriminators, which generate standardized digital pulses (NIM logic) when exceeding a set threshold. This suppresses baseline noise and ensures reliable timing signals [18].

2.5.6 Coincidence Unit (KO)

Coincidence logic—typically 2-fold or 3-fold—is implemented with fast logic circuits (e.g., CPLD, NIM modules) to identify true particle events and reject random background [2].

2.5.7 Analog to Digital Converter (ADC)

ADC modules digitize pulse amplitude, allowing energy estimation of the incident radiation. Wilkinson-type ADCs convert amplitude to digital counts via timed ramps.

2.5.8 Time to Digital Converter (TDC)

TDCs record the arrival times of START/STOP signals with 1–2 ns resolution [2], enabling timing analyses such as time-of-flight measurements.

2.5.9 Scaler (SC)

Scalers tally logic pulses over specified time intervals, providing direct measurement of detector counting rates.

2.5.10 Cable Delay (KV)

Nanosecond-scale delays are implemented by adjusting cable lengths (5 ns/m in coaxial cables) to align signal arrival times for coincidence matching [17].

3 Experimental Analysis

3.1 Properties of the signal and setting up electronic

3.1.1 Signal of photomultiplier and amplifier

The aim of this step was to study the shape and amplification of the signals coming from the photomultiplier tubes (PMTs) and their respective amplifiers. An oscilloscope was used to simultaneously display the raw and amplified signals from PMT A. Channel 1 (yellow) shows the unamplified signal, and Channel 3 (blue) displays the corresponding amplified output.

Figure 3.1 shows a typical oscilloscope capture. The unamplified PMT signal reached a minimum of -57.2 mV , while the amplified signal peaked at -572 mV . From this, the amplification factor of the amplifier (PMA) was determined as:

$$G = \frac{\text{Amplified signal}}{\text{Unamplified signal}} = \frac{572\text{ mV}}{57.2\text{ mV}} = 10$$

This gain factor remains consistent across a range of input amplitudes and confirms proper functioning of the amplifier chain.

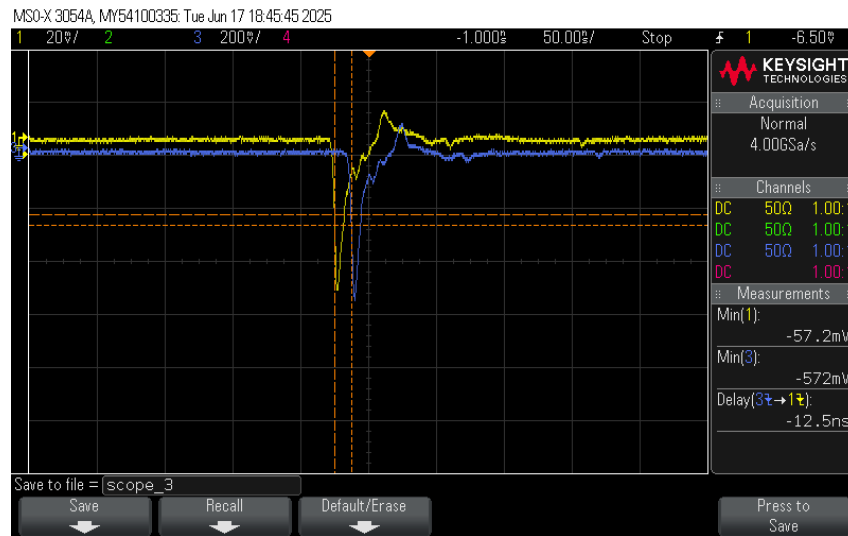


Figure 3.1: Oscilloscope display of PMT A signal: Channel 1 (yellow) is unamplified; Channel 3 (blue) is amplified.

3.1.2 Start and stop signals

To ensure correct operation of the timing and gating system, the output signals from the PMTs, discriminators, coincidence logic, and converters were observed and verified using an oscilloscope.

Figure 3.2 shows the discriminator outputs of PMT A and PMT B. Both signals exhibit identical rise and fall times, confirming temporal alignment and validating the proper coincidence condition necessary for event triggering.

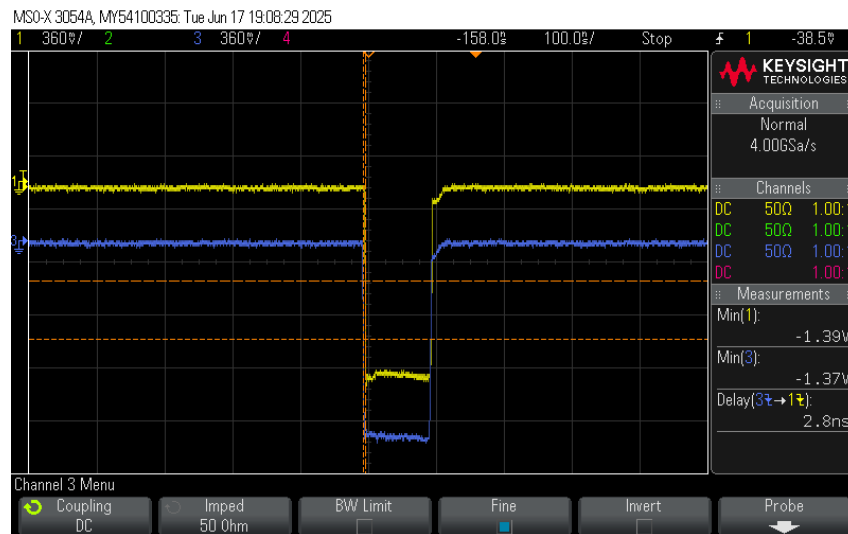


Figure 3.2: Oscilloscope display of discriminator outputs: PMT A and PMT B show temporal coincidence.

In Figure 3.3, the analog signal from the PMT (blue) is seen to be fully enclosed within the gate signal of the ADC (yellow). This confirms that the gate width is sufficient to capture the full analog pulse, ensuring accurate conversion of pulse height to digital values.

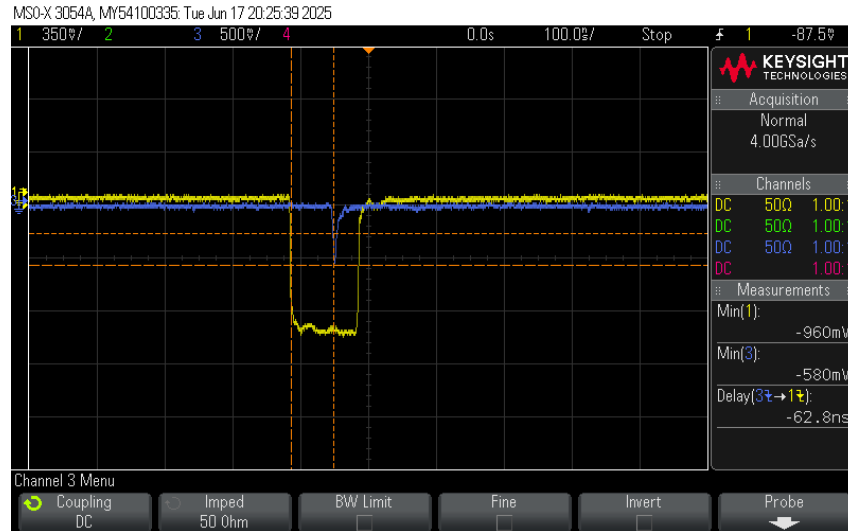


Figure 3.3: Analog signal (blue) fully enclosed within ADC gate signal (yellow), confirming proper timing for ADC integration.

Figure 3.4 displays the timing relationship between the TDC start signal and the signal from a PMT. The signal from the PMT (blue) arrives after the starting signal (yellow) at TDC.

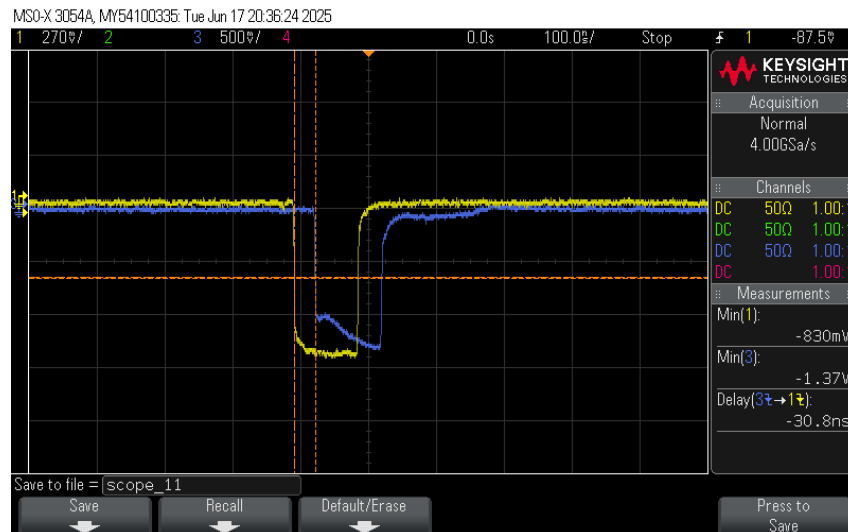


Figure 3.4: Oscilloscope display showing the TDC start signal (yellow) arriving before the PMT stop signal (blue).

3.1.3 Impact of background light

In this part of the experiment, the influence of ambient room light on the signal acquisition was studied by recording ADC spectra under two conditions: with room lights OFF and windows closed (closed room) and with room lights ON and windows open (open room). Both measurements were conducted under the same coincidence condition (PMT A), over a time interval of 180 seconds. The recorded spectra for both cases are shown in Figure 3.5 and Figure 3.6. To ensure meaningful comparison and to eliminate artifacts, only ADC channels up to 1000 were considered.

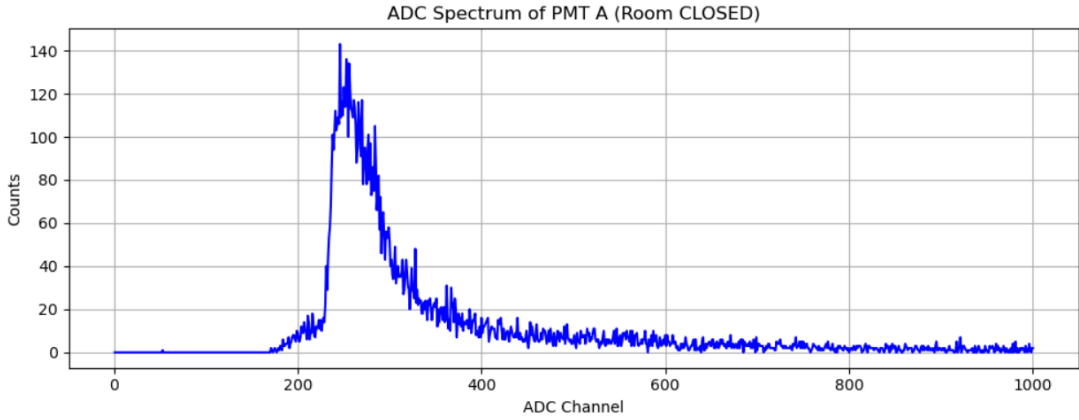


Figure 3.5: ADC spectrum of PMT A with room lights off (closed room)

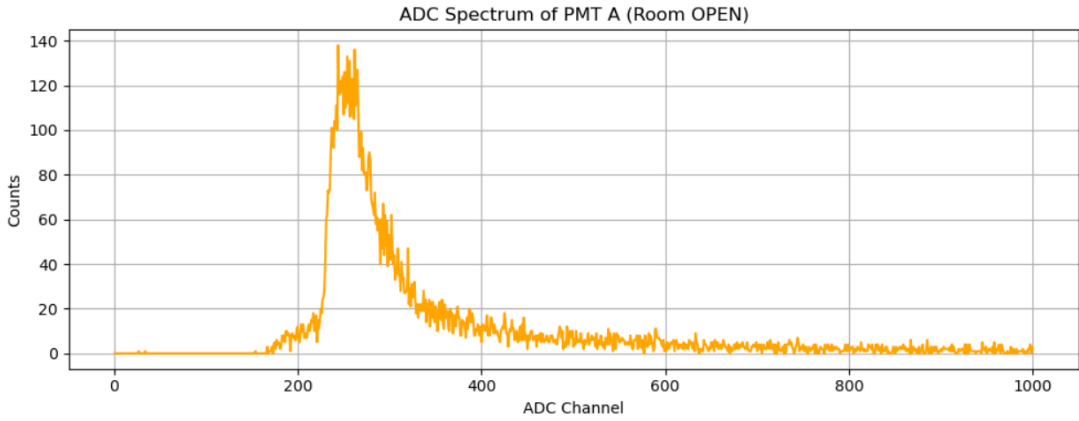


Figure 3.6: ADC spectrum of PMT A with room lights on (open room)

The shape of both spectra is similar, showing a clear peak followed by a trailing edge, corresponding to energy deposited by passing muons. No visible distortion or additional low-energy background is observed in the spectrum taken with the lights on, suggesting that the setup was not sensitive to ambient light under the used coincidence condition.

To quantify this, the total number of counts was integrated from each spectrum and used to calculate the rates. For verification, scaler measurements were also taken. These are shown in Table 3.1 and confirm that the observed effect of ambient light is negligible. The differences are minimal and lie within expected statistical fluctuations.

Table 3.1: Comparison of ADC counts, rates, and scaler counts (Channel ≤ 1000)

Condition	Total ADC Counts	Rate (counts/s)	Scaler Counts
Room Closed	11,277	62.65	15,272
Room Open	11,149	61.94	15,031

This means that the optical shielding of the scintillators and PMTs was sufficient, and the discriminator-coincidence logic reliably filtered out random signals. Since background photons typically generate isolated signals in individual detectors, they fail to meet the coincidence requirement and are not recorded.

3.1.4 Instrument calibration

In order to obtain meaningful physical quantities from the measured spectra, it is necessary to calibrate the detection system both in time and in energy.

Time calibration

To calibrate the time bins of the Time-to-Digital Converter (TDC), a series of cable delays were introduced into the signal path for each PMT. The delays used were 0 ns, 2 ns, 4 ns, 8 ns, and 16 ns. For each delay setting, TDC spectra were recorded, and the corresponding peak positions were extracted using Gaussian fits.

Representative TDC spectra for each PMT are shown in Figures 3.7–3.10. In each spectrum, five distinct peaks can be observed, corresponding to the applied cable delays. These peak positions form the basis for determining the time calibration curves.

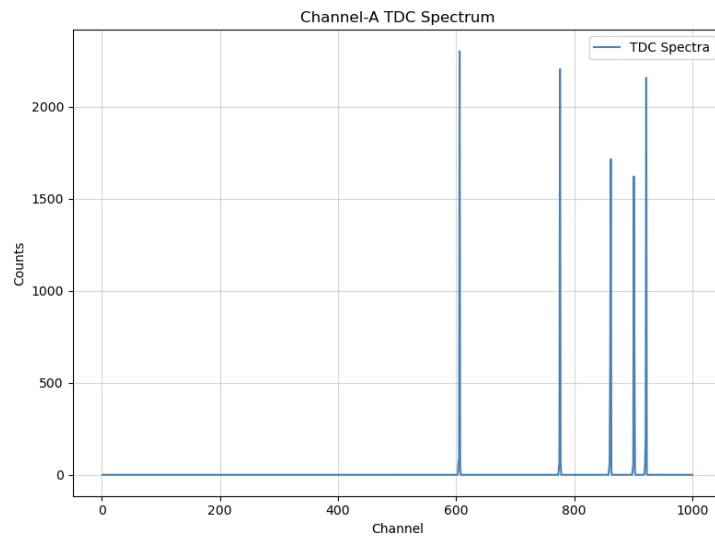


Figure 3.7: TDC spectrum of PMT A for different cable delays.

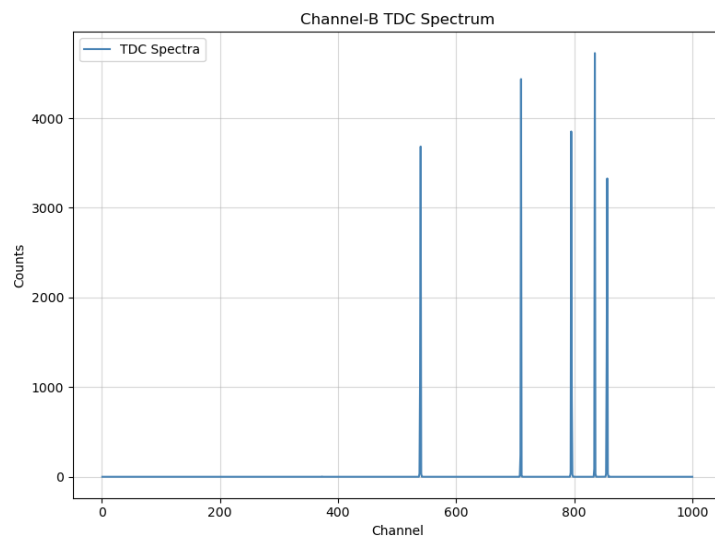


Figure 3.8: TDC spectrum of PMT B for different cable delays.

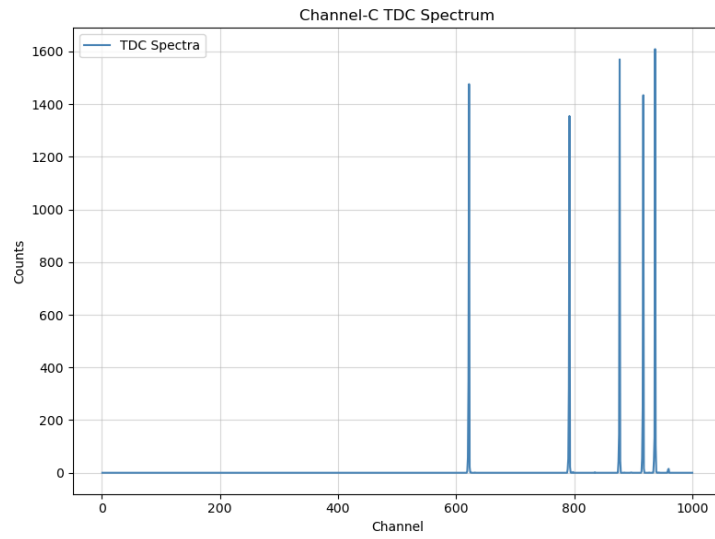


Figure 3.9: TDC spectrum of PMT C for different cable delays.

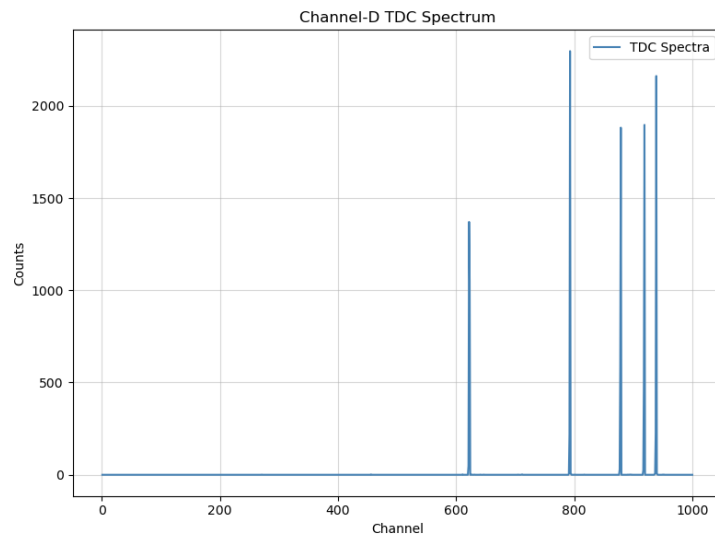


Figure 3.10: TDC spectrum of PMT D for different cable delays.

The extracted peak positions were plotted against their respective delay values, and a linear fit was performed to determine the relationship between TDC channel number and physical time in nanoseconds. The resulting calibration curves are shown in Figure 3.11, and the fit parameters are summarized in Table 3.2.

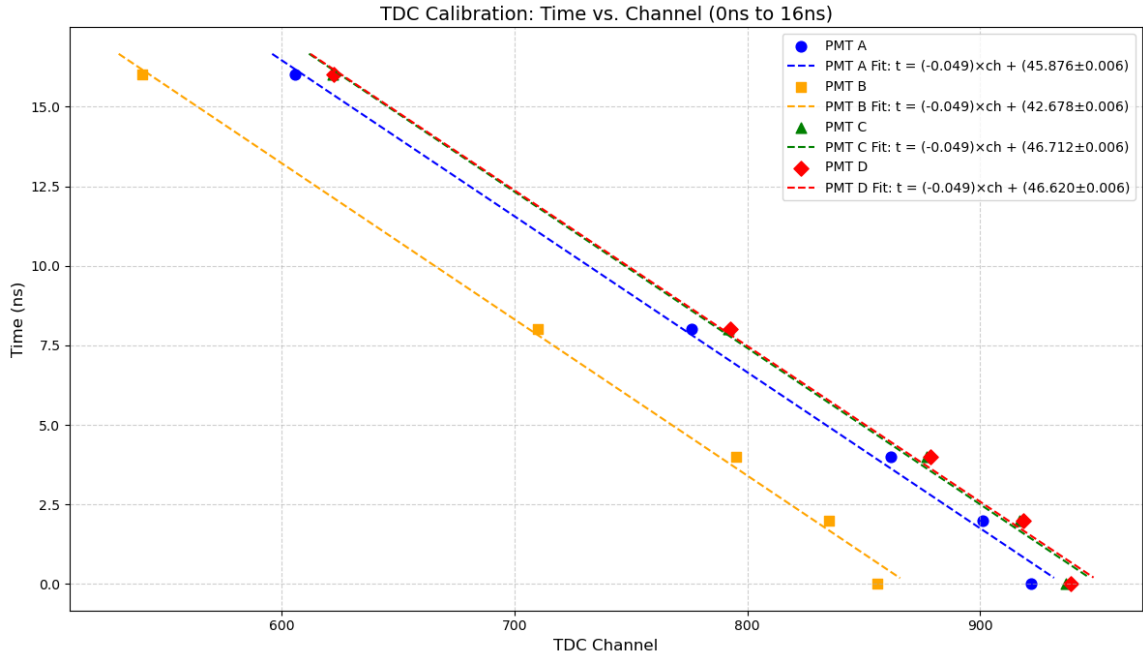


Figure 3.11: TDC calibration: Delay time versus channel number for each PMT with linear fits.

Table 3.2: Linear fit parameters for TDC time calibration

PMT	Slope m (ns/channel)	Intercept b (ns)
PMT A	-0.049	45.876 ± 0.006
PMT B	-0.049	42.678 ± 0.006
PMT C	-0.049	46.712 ± 0.006
PMT D	-0.049	46.620 ± 0.006

The nearly identical slopes indicate a consistent time resolution of approximately 49 ps/channel across all channels. The small differences in intercept values can be attributed to internal timing offsets or variations in cable lengths. These calibration functions will be used in subsequent measurements to convert TDC channels into physical time differences between PMT signals.

Energy calibration

To convert ADC channels into physical energy units (MeV), each PMT was calibrated using known gamma-ray sources: ^{60}Co and ^{22}Na . The calibration relies on identifying the Compton edge from the source spectra and the zero-energy reference point.

The zero-energy bin was determined by triggering on one channel (e.g., PMT A) and recording on a secondary channel (e.g., PMT C). This setup ensured that the recorded ADC spectrum

on the secondary channel primarily contained events where no energy was deposited, resulting in a distinct zero-energy peak. Separately, background spectra were taken without placing any radioactive source near the setup, as recommended in the manual. These spectra were used for background subtraction from the source data.

The expected Compton edge positions are 1117 keV for ^{60}Co and 1274.5 keV for ^{22}Na . After background subtraction, the Compton shoulder was fitted using the Moyal distribution, which provides a good approximation of a Landau-like energy deposition in thin scintillators [12]. The function is given by:

$$f(x) = \frac{A}{\sigma} \exp \left[-\frac{1}{2} \left(\frac{x - \mu}{\sigma} + \exp \left(-\frac{x - \mu}{\sigma} \right) \right) \right] \quad (3.1)$$

where A is the amplitude, μ the most probable value, and σ the width parameter. The zero-energy peak was fitted using a Gaussian. The extracted peak positions (in ADC channels) are summarized in Table 3.3.

Co-60 ADC Spectra (Background Subtracted) with Moyal Fit and Compton Edge

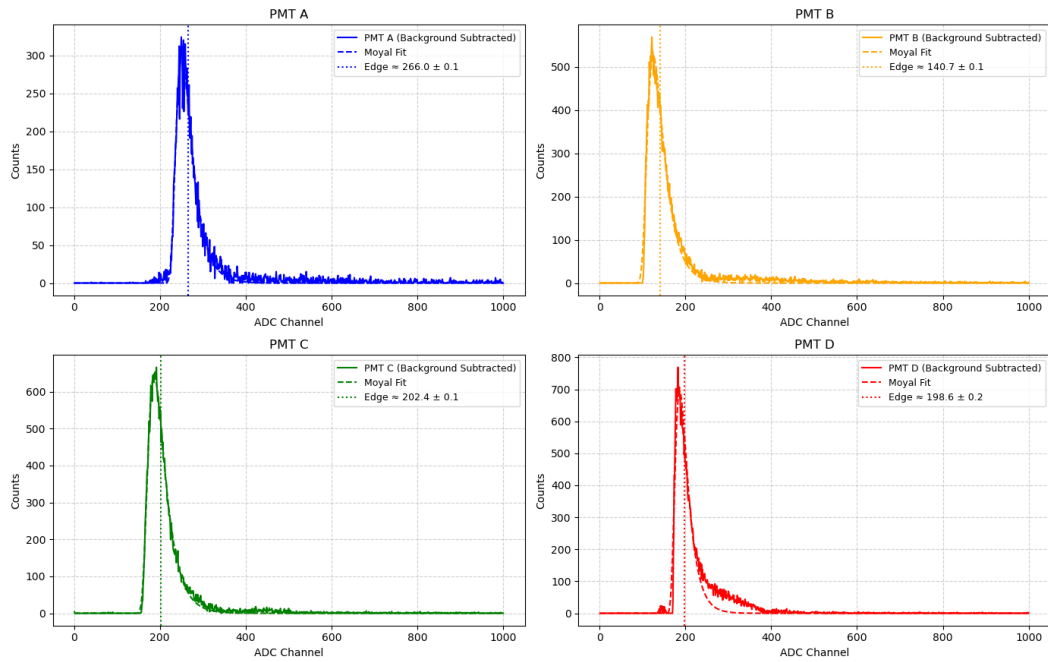


Figure 3.12: Background-subtracted Co-60 ADC spectra with Moyal fits and extracted Compton edge positions for all PMTs.

Na-22 ADC Spectra (BG Subtracted) with Moyal Fit and Zero-Energy Peak

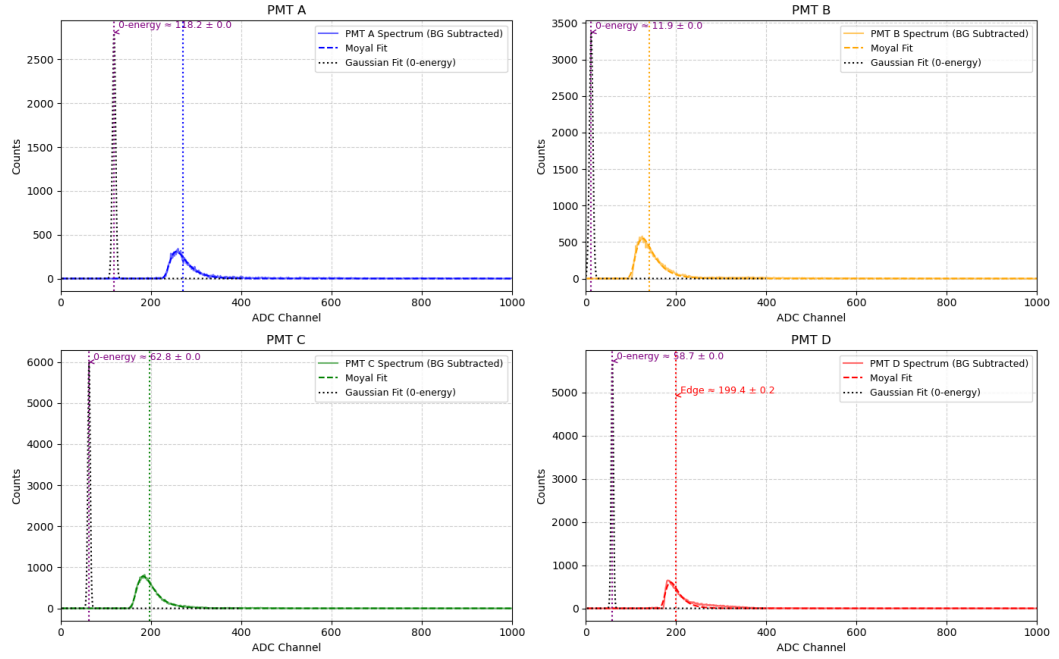


Figure 3.13: Background-subtracted Na-22 ADC spectra with Moyal and Gaussian fits. The Compton edge and zero-energy peak are marked.

Table 3.3: Zero-energy and Compton edge positions (ADC channel numbers) for each PMT

PMT	Zero-energy (ADC)	Co Compton Edge (ADC)	Na Compton Edge (ADC)
PMT A	118.2 ± 0.02	266.0 ± 0.15	271.6 ± 0.14
PMT B	11.9 ± 0.01	140.7 ± 0.13	141.2 ± 0.14
PMT C	62.8 ± 0.01	202.4 ± 0.07	197.6 ± 0.07
PMT D	58.7 ± 0.01	198.6 ± 0.18	198.5 ± 0.21

With these three reference points—zero energy, Co-60 Compton edge (1.117 MeV), and Na-22 Compton edge (1.2745 MeV)—a linear fit of the form:

$$E = m \cdot x + c \quad (3.2)$$

was performed for each PMT, where E is the energy in MeV, x the ADC channel number, and m and c are the slope and intercept of the calibration line. The fitted parameters are presented in Table 3.4, and the calibration lines are shown in Figure 3.14.

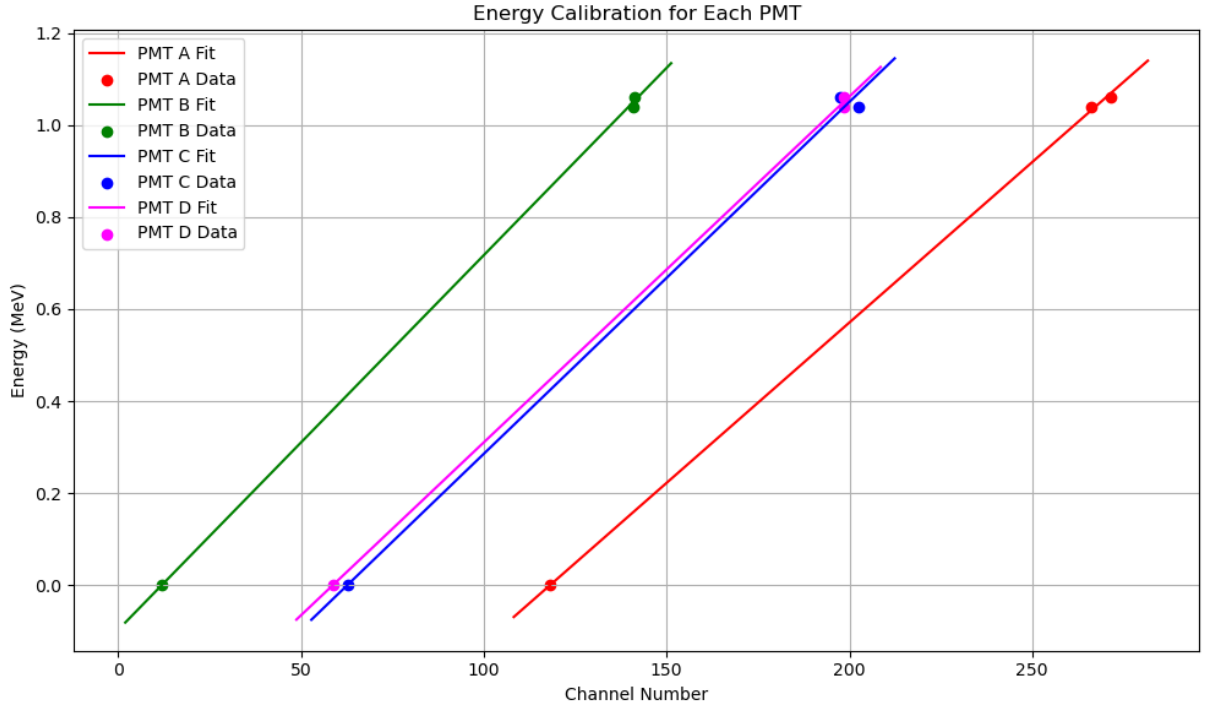


Figure 3.14: Energy calibration plot showing linear fits for each PMT.

Table 3.4: Energy calibration fit parameters for each PMT

PMT	Slope m (MeV/channel)	Intercept c (MeV)
PMT A	$(6.973 \pm 0.098) \times 10^{-3}$	-0.824 ± 0.023
PMT B	$(8.144 \pm 0.119) \times 10^{-3}$	-0.097 ± 0.014
PMT C	$(7.650 \pm 0.368) \times 10^{-3}$	-0.480 ± 0.062
PMT D	$(7.518 \pm 0.139) \times 10^{-3}$	-0.442 ± 0.023

These calibration functions will be used throughout the remaining analysis to convert ADC channels into deposited energy in MeV for each PMT.

3.1.5 Overlap Coincidence

This part of the experiment investigates how the coincidence condition affects the TDC output spectrum depending on whether signals from PMT A and PMT B arrive simultaneously or not. By adjusting the relative delays between the two channels, we explore how the overlap between discriminator pulses translates to valid coincidence events, and how the pulse width of the logic signals affects this condition.

The discriminator output has a finite width Δt , typically around 20–30 ns. Only if the signals from PMT A and PMT B overlap within this window, the logical AND (coincidence) condition

is fulfilled, generating a trigger for further data acquisition. This experiment validates this behavior by comparing three configurations:

1. Coincidence on PMT A alone (baseline).
2. Coincidence A+B without any delay.
3. Coincidence A+B with an intentional 16 ns delay introduced in PMT A.

Case 1: Coincidence A Alone (Reference)

When only PMT A is used in the coincidence condition, we observe a narrow time peak in its TDC spectrum, corresponding to the muon detection in that scintillator. PMT B, in this configuration, shows a broad timing distribution as it is not gated by any coincidence logic. These are shown in Figure 3.15.

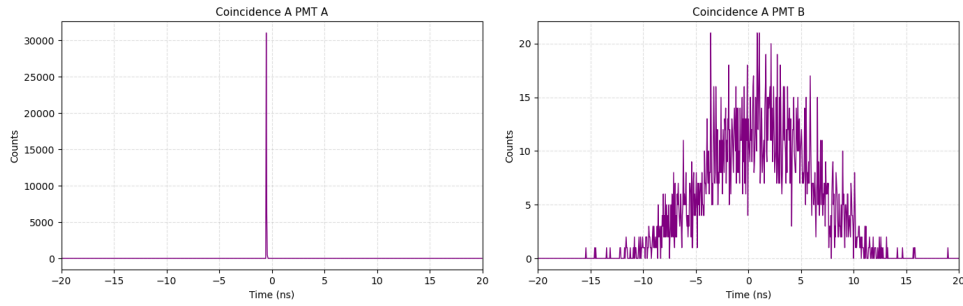


Figure 3.15: TDC spectra for PMT A and PMT B when only coincidence condition A is used. PMT A shows a sharp peak near 0 ns, while PMT B shows a broad time distribution.

Case 2: Coincidence A + B without Delay

In this configuration, both PMTs A and B must detect a signal within a small time window for the event to be accepted. Since the signals from both PMTs are (ideally) time-synchronized, valid coincidences are expected. The TDC spectra in this setup show sharp peaks for both PMTs, as illustrated in Figure 3.16.

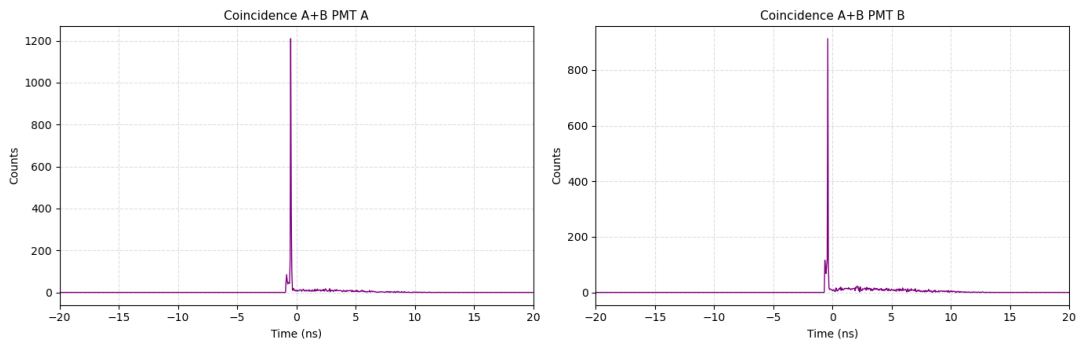


Figure 3.16: TDC spectra for PMT A and PMT B under the A+B coincidence condition with no relative delay. Both spectra now show well-defined peaks due to the successful time overlap.

Case 3: Coincidence A + B with 16 ns Delay

Here, a 16 ns delay is intentionally introduced in the signal from PMT A. This causes the discriminator pulses from A and B to no longer overlap in time, preventing the coincidence condition from being fulfilled. As a result, the TDC spectrum for PMT A still shows a peak, but PMT B returns to a broad, noise-like time spectrum, shown in Figure 3.17.

TDC Spectra: Coincidence Condition A+B with 16ns Delay

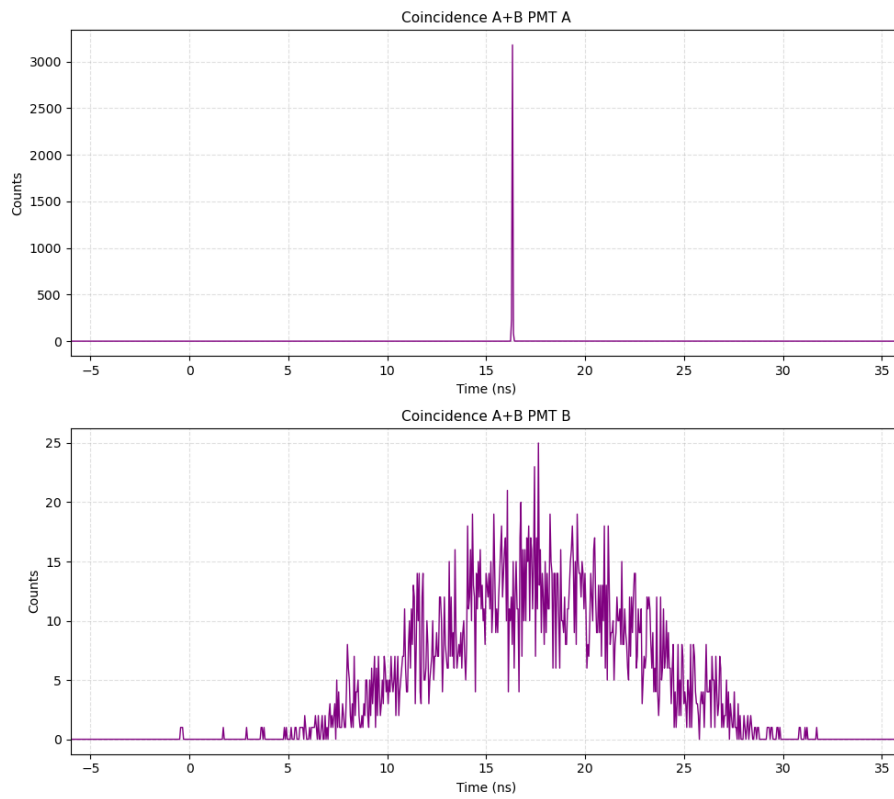


Figure 3.17: TDC spectra with a 16 ns delay in PMT A. The peak in PMT A remains, but PMT B loses its sharp timing response due to lack of overlapping signals.

2D Coincidence Histograms

The effect of coincidence logic is further visualized using 2D histograms that map the TDC channel of PMT A versus PMT B. In the case without delay (Figure 3.18), we observe a sharp vertical band indicating correlated events. When delay is introduced (Figure 3.19), this alignment is disrupted, and the vertical band becomes sparse or disappears altogether. A reference histogram for coincidence A only is shown in Figure 3.20.

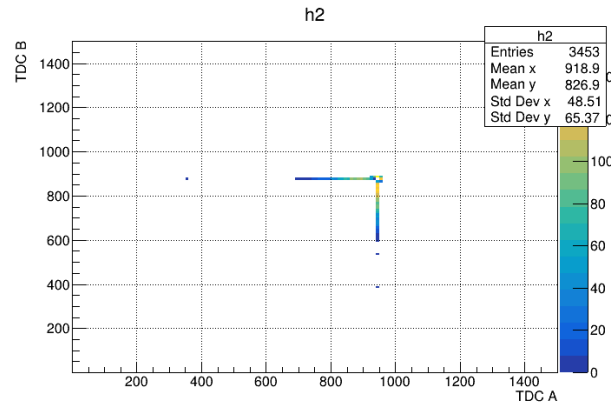


Figure 3.18: 2D TDC histogram with A+B coincidence and no delay. A narrow vertical band appears indicating tight timing correlation between A and B.

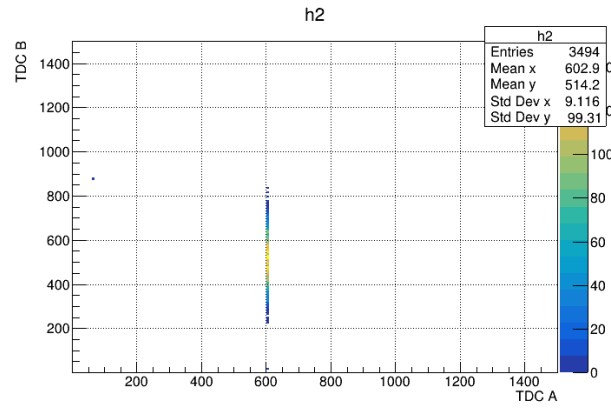


Figure 3.19: 2D TDC histogram with A+B coincidence and 16 ns delay. The previously seen vertical band becomes diffuse, confirming the loss of valid overlaps.

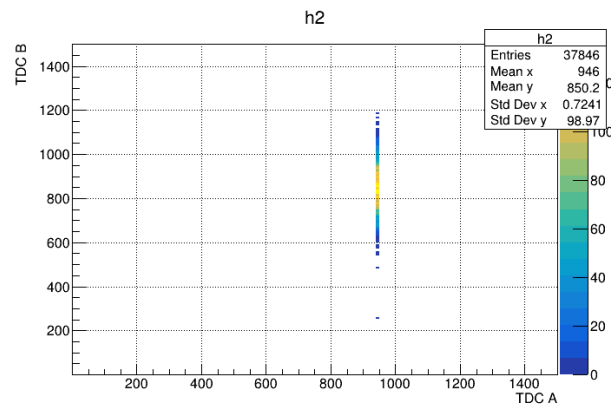


Figure 3.20: 2D histogram for coincidence A only (PMT A triggered). PMT B is not correlated, and no clear band is seen.

Conclusion

This section demonstrates that successful coincidence triggering in the experiment depends critically on the timing overlap between discriminator outputs. By shifting one signal outside the effective pulse width, coincidence efficiency drops significantly.

This behavior validates the overlap logic of the DAQ system and highlights the importance of carefully tuned delays in time-sensitive coincidence experiments.

3.2 Detector performance with ^{60}Co -source

This part of the experiment aims to obtain additional information about the detector using the ^{60}Co -source.

3.2.1 Attenuation in the Scintillator

This part investigates the attenuation of scintillation light within the plastic scintillator by placing a Co-60 source at three different lateral positions: -25 cm , 0 cm , and $+25\text{ cm}$ along the detector length. ADC spectra were acquired from both PMT A and PMT B for each of these configurations under the coincidence condition A. All spectra were background-subtracted using data taken under identical acquisition settings but without any source, and subsequently energy-calibrated based on earlier calibration constants.

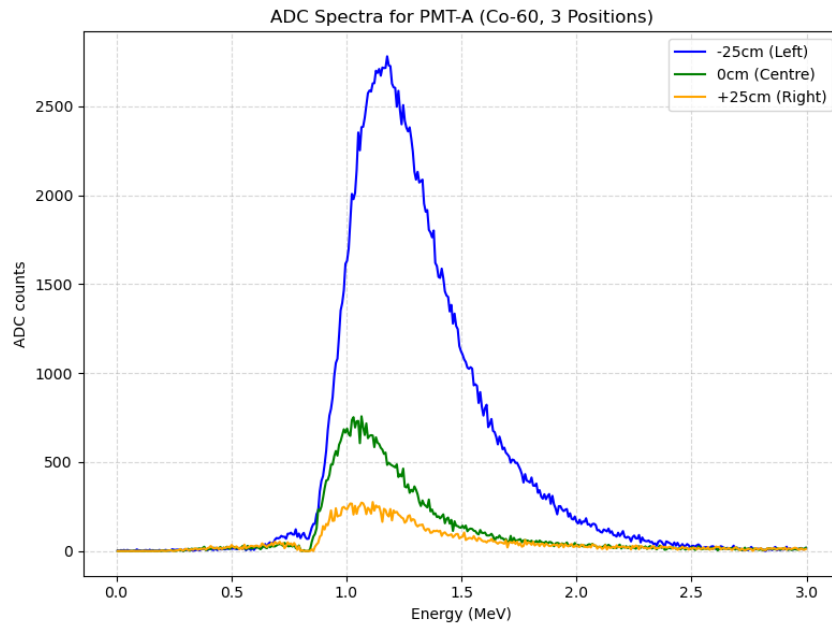


Figure 3.21: ADC spectra for PMT-A with Co-60 at three positions along the scintillator.

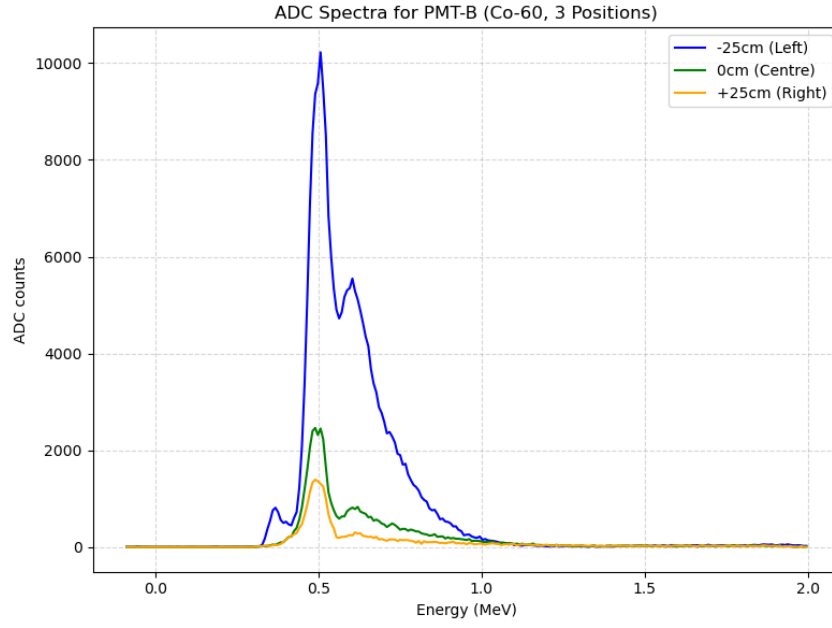


Figure 3.22: ADC spectra for PMT-B with Co-60 at three positions along the scintillator.

As seen in the plots above, the ADC peak heights decrease as the source is moved farther from the respective PMT. This confirms the expected attenuation of light as it propagates through the scintillator.

To quantify this attenuation, the total number of background-subtracted ADC counts for each position was integrated. These counts were then plotted as a function of source distance from the respective PMT. The attenuation model is given by:

$$N(x) = A_0 e^{-\mu x} \quad (3.3)$$

where $N(x)$ is the observed count at distance x , A_0 is the amplitude at $x = 0$, and μ is the attenuation coefficient. The fit was performed using nonlinear least squares, and the extracted values of μ are shown in the plots below for each PMT.

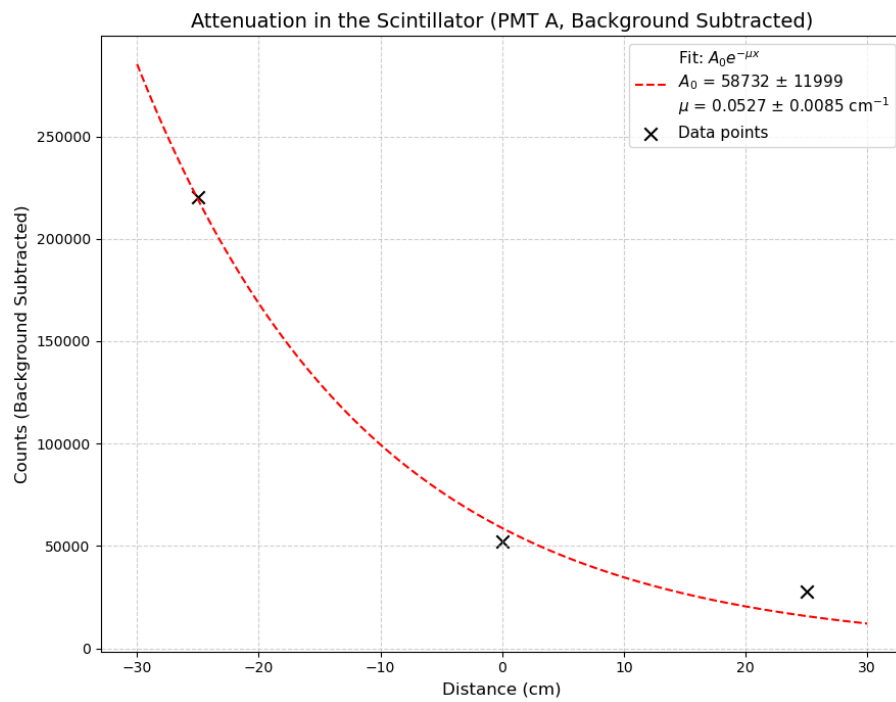


Figure 3.23: Attenuation curve for PMT A. The exponential fit yields $\mu = 0.0527 \pm 0.0085 \text{ cm}^{-1}$.

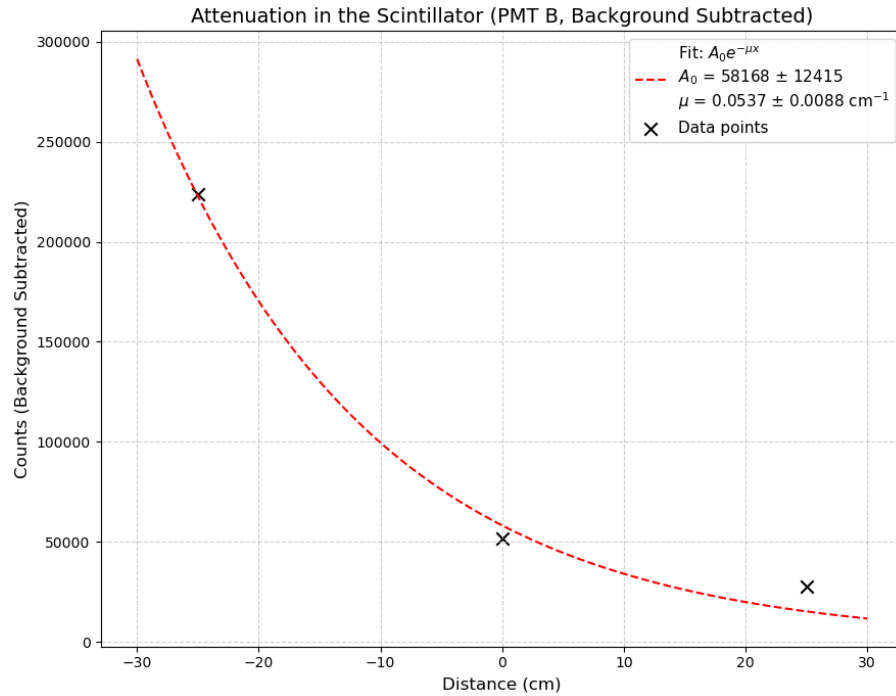


Figure 3.24: Attenuation curve for PMT B. The exponential fit yields $\mu = 0.0537 \pm 0.0088 \text{ cm}^{-1}$.

Table 3.5: Fitted parameters from attenuation model $N(x) = A_0 e^{-\mu x}$ for PMT A and PMT B

PMT	A_0 (counts)	μ (cm^{-1})
PMT A	58732 ± 11999	0.0527 ± 0.0085
PMT B	58168 ± 12415	0.0537 ± 0.0088

The extracted attenuation coefficients for both PMTs are consistent within errors, with values around 0.053 cm^{-1} , indicating symmetric light loss within the scintillator material. The results confirm the expected exponential dependence of light attenuation and validate the alignment and optical coupling of the PMTs.

3.2.2 Free vs Coincidence Spectra

In this part, we compare the ADC spectra of two scenarios: the *free spectra*, where only a single coincidence condition (in this case, A) is used to trigger the acquisition, and the *coincidence spectra*, where both PMT-A and PMT-B must detect an event simultaneously (coincidence A+B). These measurements help distinguish genuine coincident events from background and uncorrelated detections.

The measurements were taken at the central position of the scintillator bar (0 cm). Four spectra were recorded for PMT-A and PMT-B for both coincidence conditions A and A+B

All spectra were first background subtracted using data collected without any radioactive source, and then converted from ADC channel to energy using the calibration functions:

$$E_A(\text{MeV}) = 0.006973 \cdot \text{ADC} - 0.823878 \quad (3.4)$$

$$E_B(\text{MeV}) = 0.008144 \cdot \text{ADC} - 0.097116 \quad (3.5)$$

The resulting energy spectra are shown in Figure 3.25, highlighting a significant suppression of background in the coincidence spectra, especially in the low-energy region. This is expected, as requiring both PMTs to detect a particle simultaneously reduces the chance of random noise or uncorrelated photons being registered as valid events.

Table 3.6: Total event counts and derived count rates for each PMT under Free and Coincidence conditions.

Condition	Total Counts	Count Rate (Hz)
PMT-A Free	49547	117.97
PMT-B Free	50705	120.73
PMT-A Coincident	1555	3.70
PMT-B Coincident	2355	5.61

From Table 3.6, we observe a sharp drop in both total counts and count rates when going from the free to the coincidence mode, demonstrating the effectiveness of coincidence logic in filtering out background and isolating correlated γ events. PMT-B shows a slightly higher rate than PMT-A, which can be attributed to subtle differences in gain and positioning.

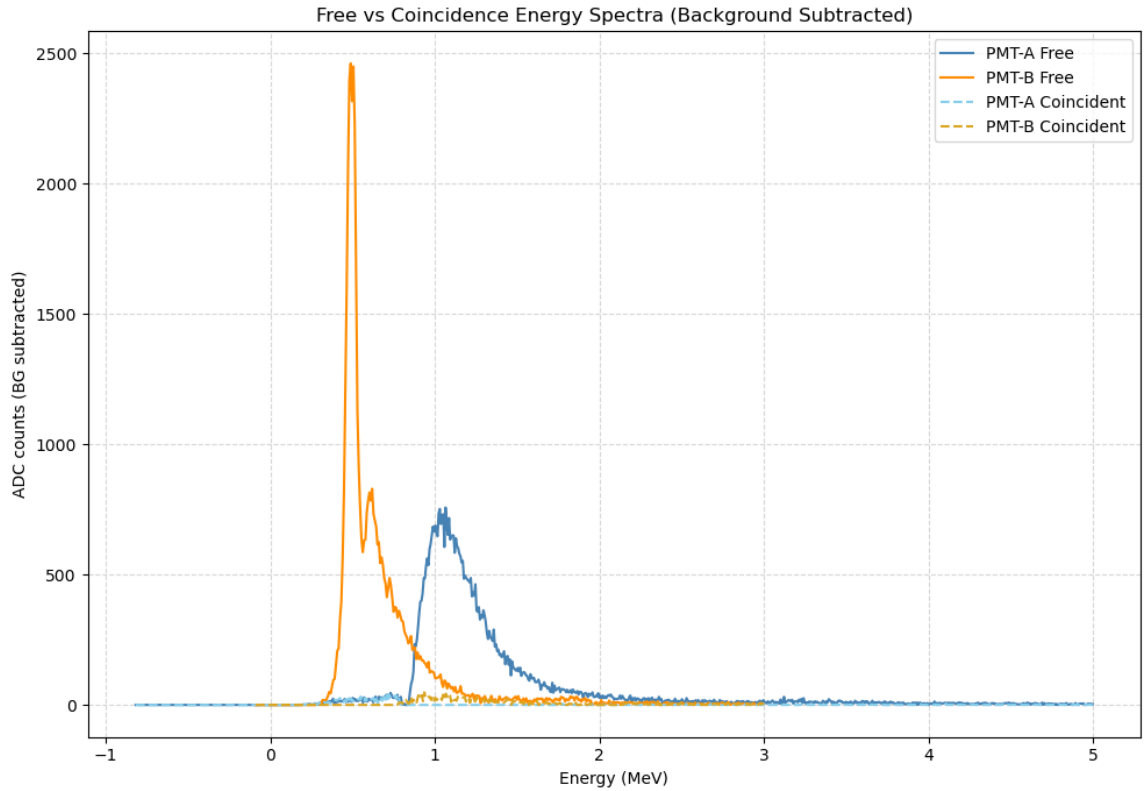


Figure 3.25: Comparison of background-subtracted ADC spectra for PMT-A and PMT-B under Free and Coincidence (A+B) conditions.

3.2.3 Speed of Light in the Scintillator

To determine the effective speed of light in the scintillator material, we positioned a ^{60}Co source at three different locations: left (-25 cm), center (0 cm), and right ($+25\text{ cm}$) on the detector.

For each position, we recorded the TDC spectra of PMT A and PMT B under coincidence condition (A+B). However, the spectra obtained from PMT A are not giving physically good results hence we used the data for just PMT B in this part and extracted the mean value μ of the time distribution using a Gaussian fit. The calibrated time spectra are shown in Figure 3.26.

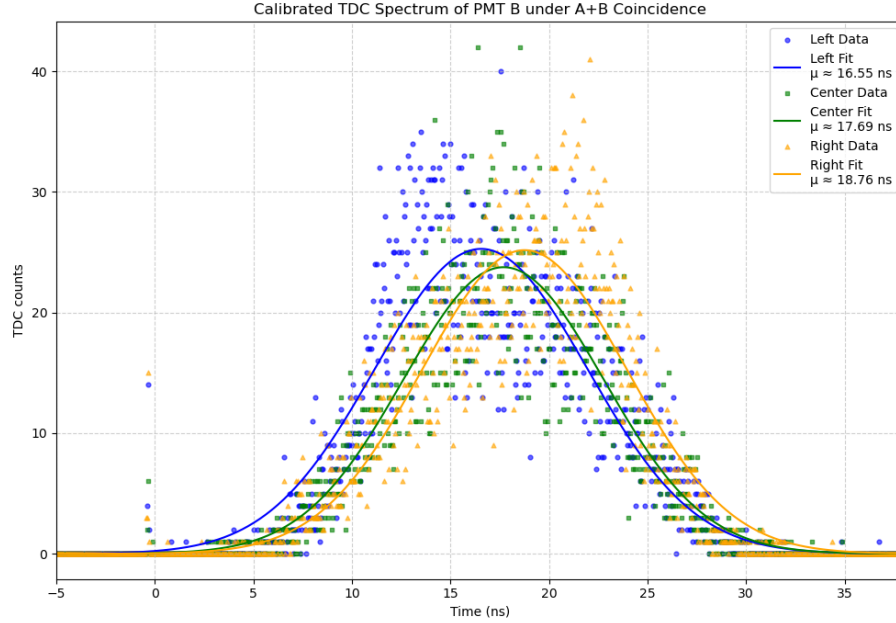


Figure 3.26: TDC spectra of PMT B for three source positions under (A+B) coincidence condition. Gaussian fits yield the μ values used for speed estimation.

- $\mu_{\text{left}} \approx 16.55 \text{ ns}$
- $\mu_{\text{center}} \approx 17.69 \text{ ns}$
- $\mu_{\text{right}} \approx 18.76 \text{ ns}$

These values were plotted against the respective distances and fitted with a linear function of the form:

$$\mu(x) = a \cdot x + b \quad (3.6)$$

where x is the source position in centimeters. The slope of the fit was found to be:

$$a = 0.0442 \text{ ns/cm} \quad (3.7)$$

The linear fit of the extracted μ values are shown in Figure 3.27.

This corresponds to an effective light speed in the scintillator:

$$c_{\text{eff}} = \frac{1}{a} = \frac{1}{0.0442 \text{ ns/cm}} = 2.26 \times 10^8 \text{ m/s} \quad (3.8)$$

From this, we calculated the refractive index of the scintillator material as:

$$n = \frac{c}{c_{\text{eff}}} = \frac{3.00 \times 10^8}{2.26 \times 10^8} \approx 1.33 \quad (3.9)$$

This result is slightly lower than the given expected value for plastic scintillator ($n \approx 1.58$), which may be due to uncertainties in the TDC calibration or the finite resolution of the timing electronics.

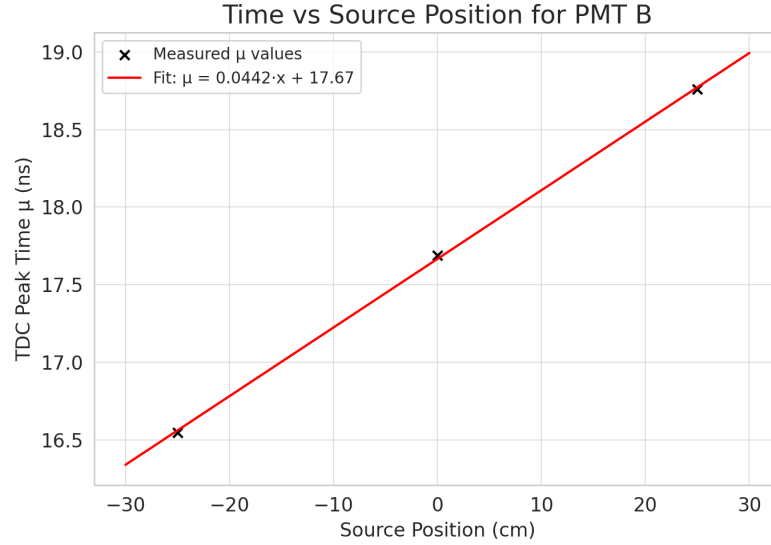


Figure 3.27: Linear fit of time-of-flight μ values as a function of source position for PMT B. The slope of the fit corresponds to an effective light speed in the scintillator.

3.3 Properties of atmospheric muons

3.3.1 Detector Resolution

To determine the time and spatial resolution of the experimental setup, we placed both scintillator plates directly on top of each other as shown in figure 3.28. Using cosmic muons passing vertically through the stacked detectors, we recorded the time-of-flight difference spectrum using PMT C under the (A+C) coincidence condition as shown in figure 3.29.

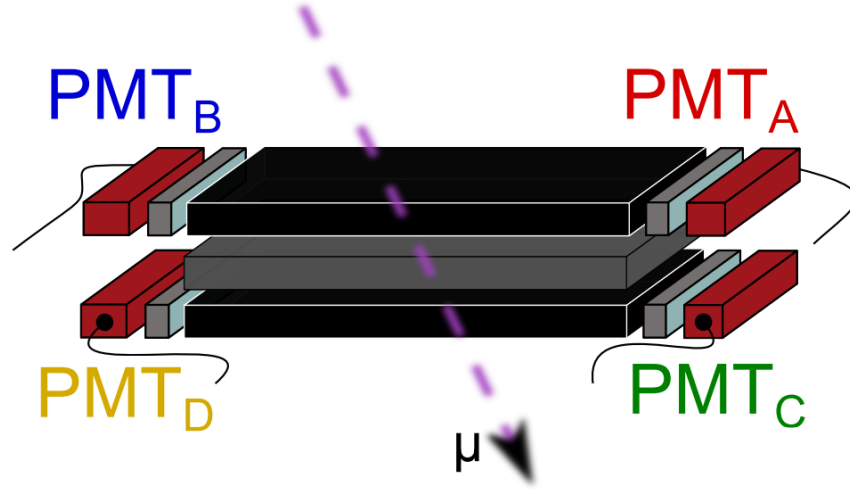


Figure 3.28: Measurement setup with detector plates on top of each other

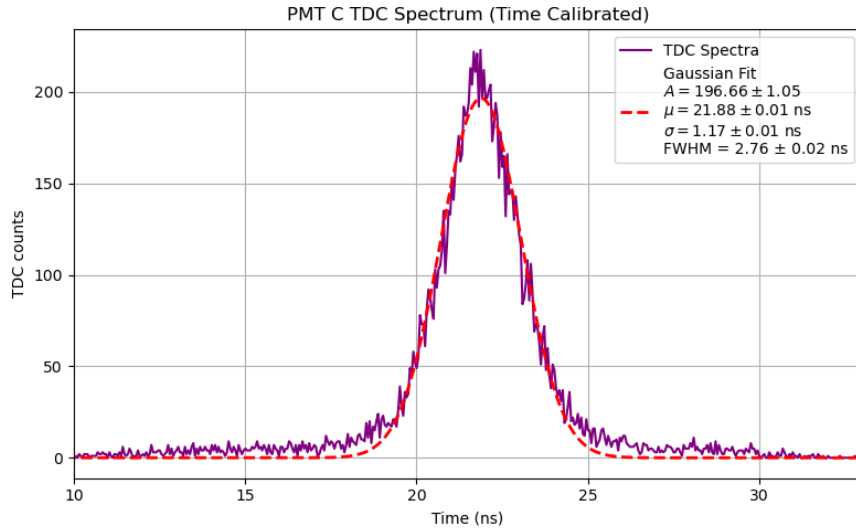


Figure 3.29: TDC spectrum of PMT C for vertically traversing muons.

The Gaussian fit that was applied to the TDC time distribution, yields the following parameters:

- Mean time $\mu = 21.88 \pm 0.01 \text{ ns}$
- Standard deviation $\sigma = 1.17 \pm 0.01 \text{ ns}$
- Full Width at Half Maximum (FWHM):

$$\text{FWHM} = 2.76 \pm 0.02 \text{ ns} \quad (3.10)$$

This FWHM defines the time resolution of the setup. To estimate the corresponding spatial resolution, we use the relation:

$$\sigma_x = \frac{c_{\text{eff}} \cdot \text{FWHM}}{2} \quad (3.11)$$

Substituting the effective speed of light in the scintillator ($c_{\text{eff}} = 2.26 \times 10^8 \text{ m/s}$) from Section 3.2.3:

$$\sigma_x = \frac{2.26 \times 10^8 \cdot 2.76 \times 10^{-9}}{2} \approx 0.312 \text{ m} = 31.2 \text{ cm} \quad (3.12)$$

This implies that the detector has a spatial resolution of approximately 31 cm, compared to the total width of the detector approximately around 34cm, meaning we cannot reliably distinguish whether a muon hit the left or right half of the scintillator.

3.3.2 Detector Preparation

To verify the detector's ability to properly record vertical muons, we acquired ADC spectra from PMT A (top plate) and PMT C (bottom plate) with the detectors stacked on top of each other under coincidence condition (A+C). Both spectra were calibrated to energy using the known calibration factors.

A Moyal distribution was fitted to each ADC spectrum. The resulting most probable energies (μ) were:

- PMT A: $\mu_A = 1.09 \pm 0.01 \text{ MeV}$
- PMT C: $\mu_C = 1.42 \pm 0.01 \text{ MeV}$

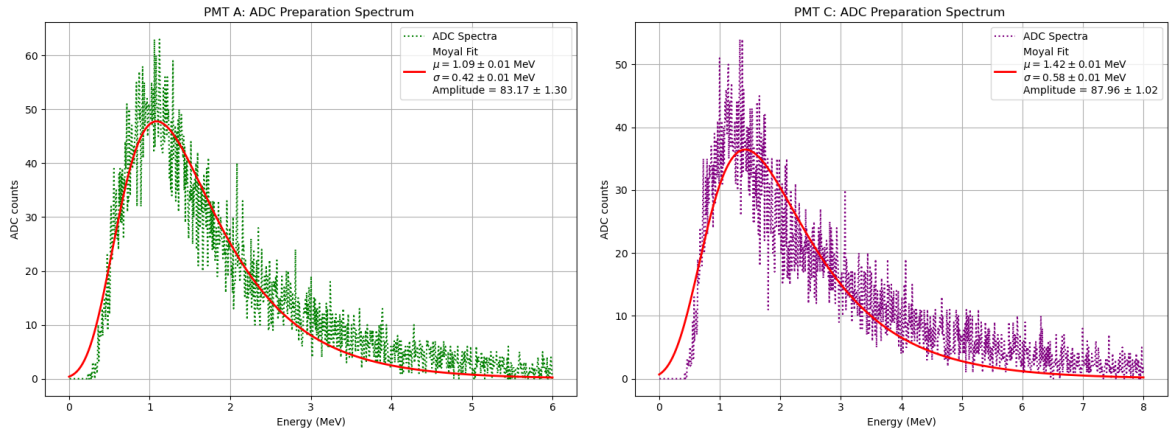


Figure 3.30: ADC energy spectra for PMT A (left) and PMT C (right) under vertical muon incidence.

The PMT C spectrum shows a slightly higher mean energy. This is expected as muons may deposit more energy while exiting through the second scintillator due to minor scattering or

angular spread. The general similarity of both spectra confirms that both detectors are properly recording muon signals.

In addition, the correlation between ADC signals of both PMTs was examined:

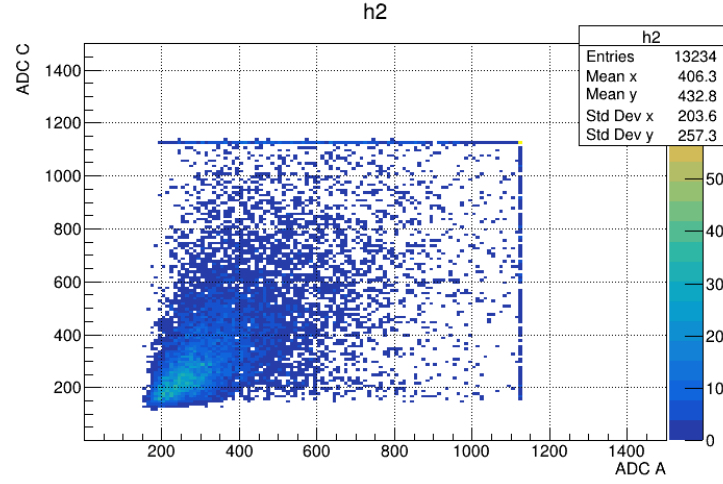


Figure 3.31: 2D scatter plot of ADC values from PMT A vs PMT C. A clear diagonal structure confirms correlated energy deposition by the same muon in both detectors.

This plot shows a strong diagonal band, confirming that the same muon deposits energy in both scintillators.

Finally, a correlation between energy deposition in PMT C and its corresponding TDC time was observed:

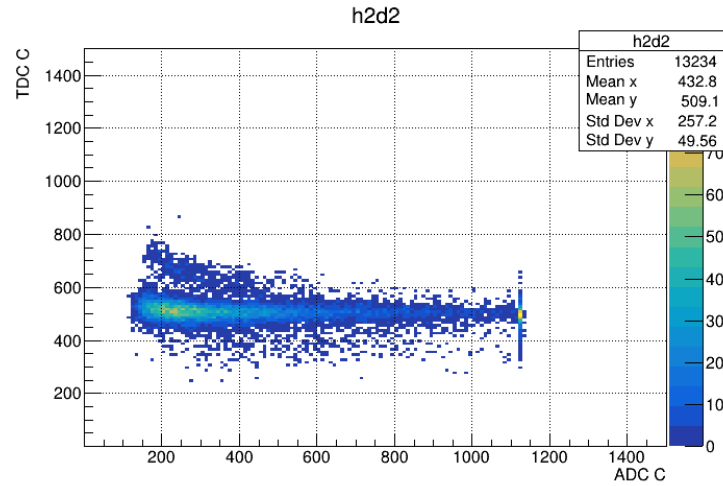


Figure 3.32: 2D scatter plot of ADC vs TDC values for PMT C. The clustering indicates consistent muon timing with varying energy loss.

Overall, the observed consistency between both ADC spectra and the clear correlations in the 2D plots validate that the detector system is properly prepared for further cosmic muon measurements.

3.3.3 Intensity and Energy Loss

In this part of the experiment, the two scintillator detectors were placed in a crossed configuration as shown in figure 3.33. This setup increases the path length for muons passing through both detectors, which should lead to a higher energy deposition compared to the vertically stacked configuration.

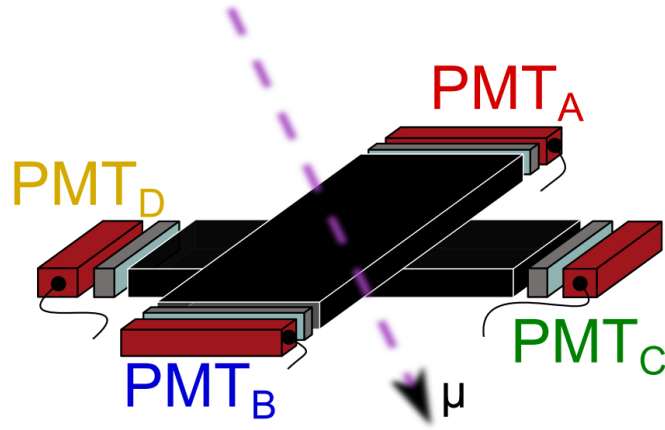


Figure 3.33: Measurement setup with crossed detector plates.

To observe this effect, we recorded ADC spectra for PMT A and PMT C in both configurations. The spectra were energy-calibrated and fitted with Moyal distributions to extract the most probable energy loss.

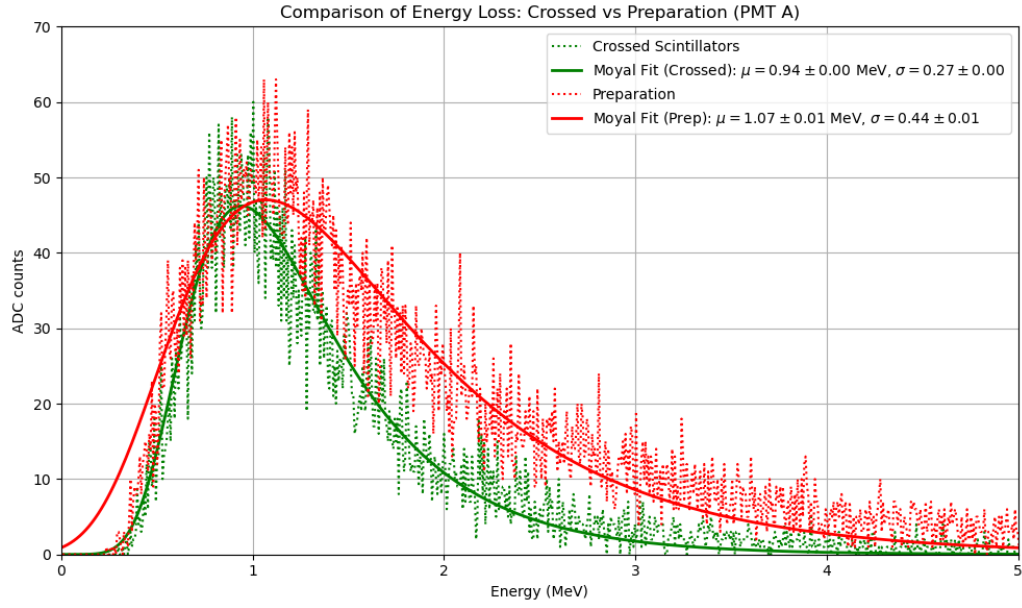


Figure 3.34: Comparison of energy loss for PMT A between the preparation and crossed configurations. The Moyal peak is slightly lower in the crossed case due to angular muons grazing the detector.

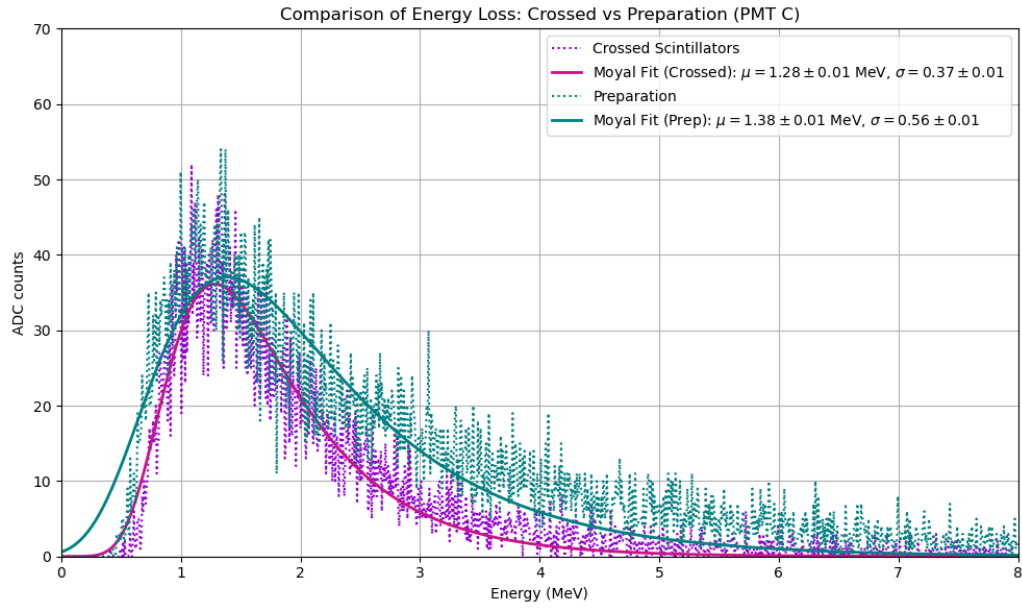


Figure 3.35: Comparison of energy loss for PMT C between the preparation and crossed configurations. The spectra show a modest increase in most probable energy in the preparation setup, with broader spread.

The extracted values are summarized below:

PMT	Setup	Most Probable Energy (μ)	σ
A	Preparation	1.07 ± 0.01 MeV	0.44 ± 0.01
A	Crossed	0.94 ± 0.00 MeV	0.27 ± 0.00
C	Preparation	1.38 ± 0.01 MeV	0.56 ± 0.01
C	Crossed	1.28 ± 0.01 MeV	0.37 ± 0.01

Table 3.7: Fitted Moyal parameters for PMT A and PMT C in both preparation and crossed configurations.

While one might expect a larger energy deposition in the crossed setup due to increased path length, the spectra for the crossed configuration actually show slightly *lower* most probable energies. This is likely because the crossed geometry allows a wider range of muon angles, including more glancing paths, which do not necessarily deposit more energy.

To estimate the muon intensity, we use the total number of recorded events in the crossed configuration:

- Total counts (PMT C): 8047
- Measurement time: 900 seconds (15 minutes)
- Active detector area: $34 \text{ cm} \times 34 \text{ cm} = 0.1156 \text{ m}^2$

The intensity is then given by:

$$I = \frac{\text{Counts}}{\text{Area} \times \text{Time}} = \frac{8047}{0.1156 \times 900} \approx 77.34 \text{ counts/m}^2.\text{s} \quad (3.13)$$

This value is consistent with the expected order of magnitude for cosmic muon flux at sea level, which is approximately $100 \text{ muons/m}^2.\text{s}$ [10], confirming the reliability of the setup and method.

3.3.4 Electromagnetic and muonic component

To examine the effect of lead shielding on cosmic muon energy deposition, we placed lead blocks between the two scintillator detectors and recorded ADC spectra from PMT A (top) and PMT C (bottom) under the same coincidence condition as before. The measurement duration remained 15 minutes.

The ADC spectra for both PMTs were energy-calibrated and fitted with Moyal distributions. A comparison with the corresponding spectra from Section 3.3.3 (without lead) is shown below.

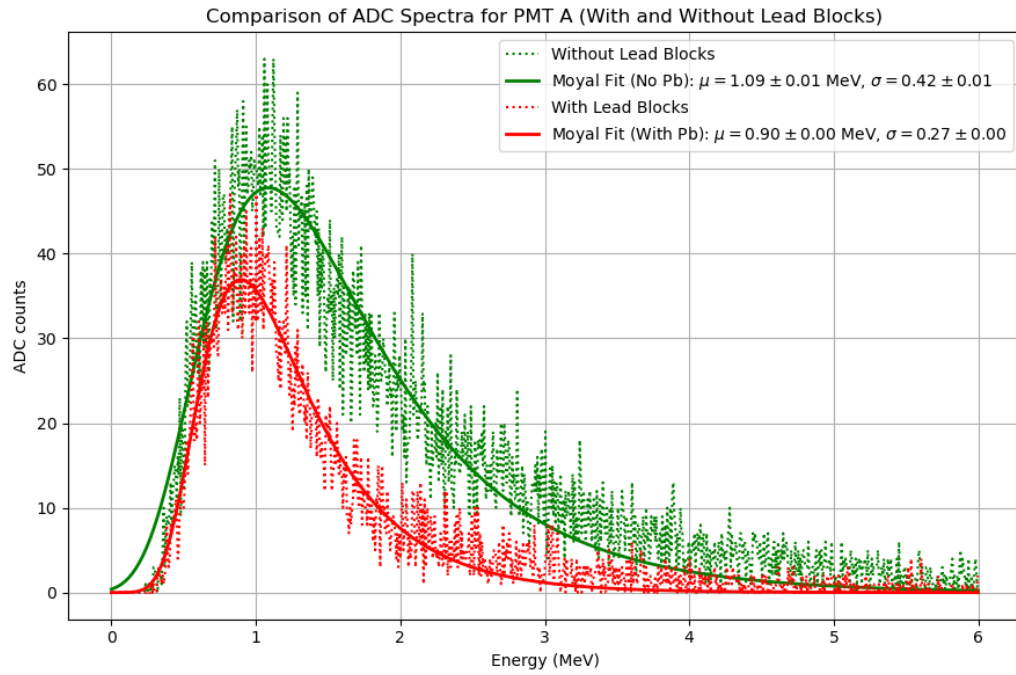


Figure 3.36: ADC spectra for PMT A with and without lead shielding. A clear suppression of the high-energy tail is visible in the lead configuration.

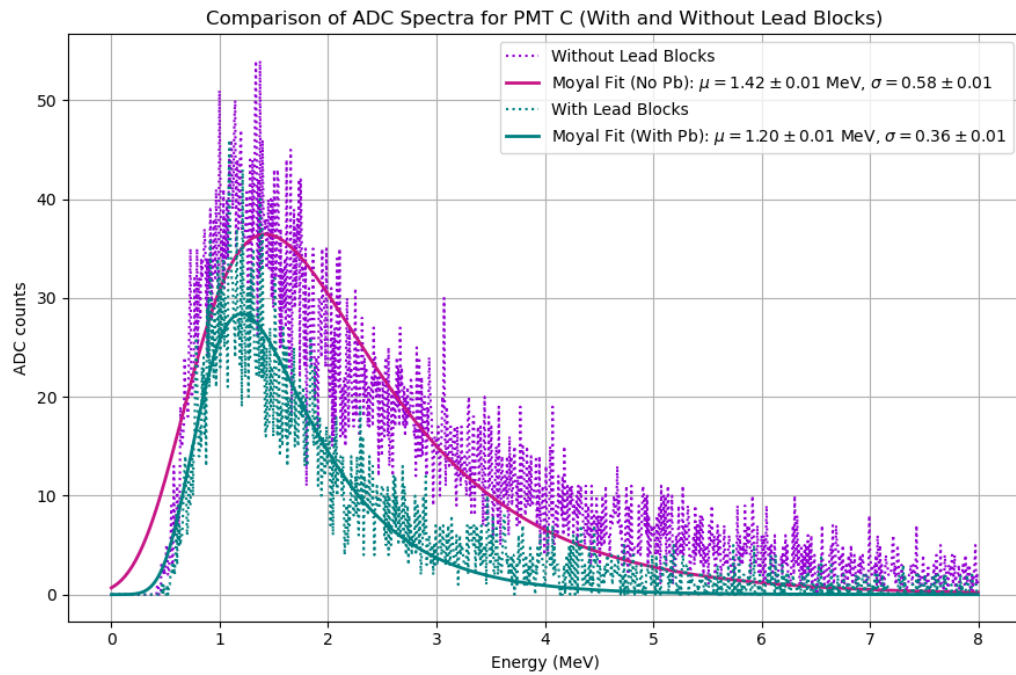


Figure 3.37: ADC spectra for PMT C with and without lead shielding. The lead absorber causes a shift in the peak position and changes the distribution shape.

The most probable energies and widths obtained from the Moyal fits are summarized below:

PMT	Lead Status	Most Probable Energy (μ)	σ
A	Without Lead	1.09 ± 0.01 MeV	0.42 ± 0.01
A	With Lead	0.90 ± 0.00 MeV	0.27 ± 0.00
C	Without Lead	1.42 ± 0.01 MeV	0.58 ± 0.01
C	With Lead	1.20 ± 0.01 MeV	0.36 ± 0.01

Table 3.8: Moyal fit results for ADC energy spectra with and without lead shielding.

The results indicate that the presence of a lead block reduces the overall energy deposition, as seen by a lower most probable value (μ) and a narrower spread (σ). This suggests that the lead layer filters out part of the particle flux, likely absorbing low-energy components or scattering muons before reaching the lower detector.

3.3.5 Speed and Zenith Angle Dependence of Muons

In this section, we investigated how the time-of-flight (TOF) and count rate of cosmic muons depend on the zenith angle θ . This was done in two steps:

1. **TOF Measurement:** The TDC spectrum of PMT C was recorded for muons entering at $\theta = 0^\circ$ (top-down) and $\theta = 180^\circ$ (bottom-up), under the coincidence condition (A+C). The path length for vertically downward muons is shorter compared to muons entering from below, which must pass through more material and traverse a longer path.
2. **Zenith Angle Flux Dependence:** The muon flux was recorded at various zenith angles from 0° to 180° by tilting the detector setup as shown in figure 3.38. Count rates were extracted from the TDC spectra by integrating over the signal window.

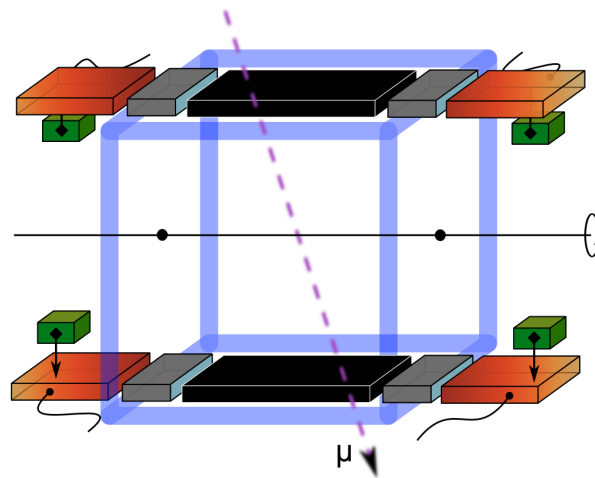


Figure 3.38: Measurement setup mounted on the rotating experiment setup.

TOF Spectrum at 0° and 180°

The time-calibrated TDC spectra for PMT C at 0° and 180° are shown in Figure 3.39. Gaussian fits were performed to extract the peak positions μ , which represent the average arrival time difference due to the different path lengths.

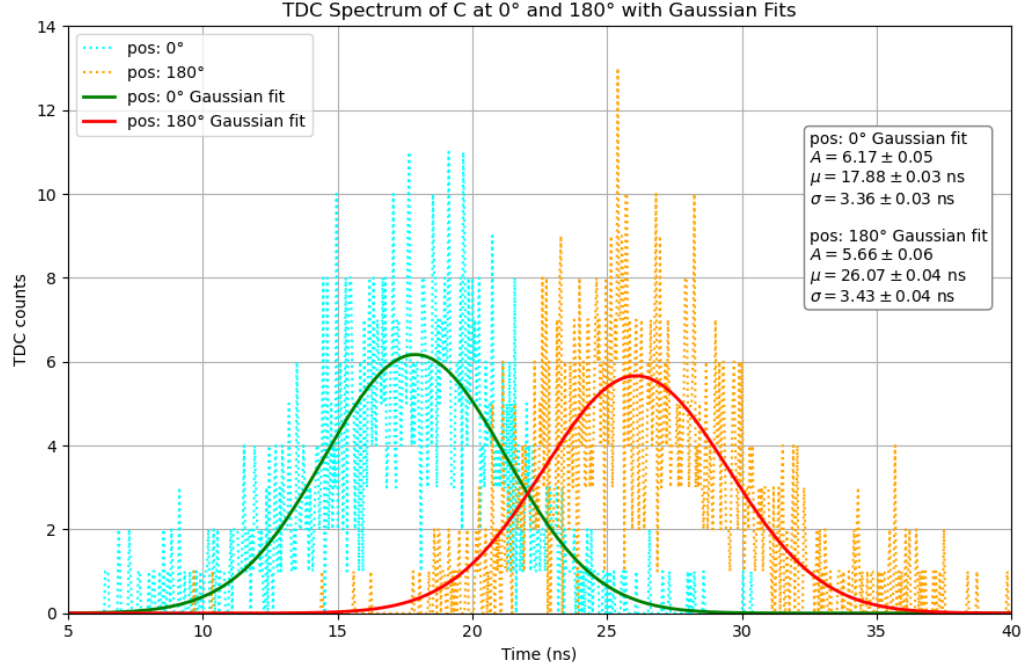


Figure 3.39: Time-of-flight spectrum of PMT C for muons arriving at 0° and 180° . A clear delay is seen for upward-going muons.

The fit results are:

- At $\theta = 0^\circ$: $\mu = 17.88 \pm 0.03$ ns
- At $\theta = 180^\circ$: $\mu = 26.07 \pm 0.04$ ns
- Time delay: $\Delta t = (8.19 \pm 0.05)$ ns

This shift in arrival time matches expectations: upward-going muons traverse a longer path, leading to increased TOF.

Zenith Angle Dependence of Muon Flux

To study how the flux of muons varies with incident angle, the count rate (in counts/minute) was extracted for various angles by integrating the TDC spectra of PMT A within a time window of 15 ns to 21 ns.

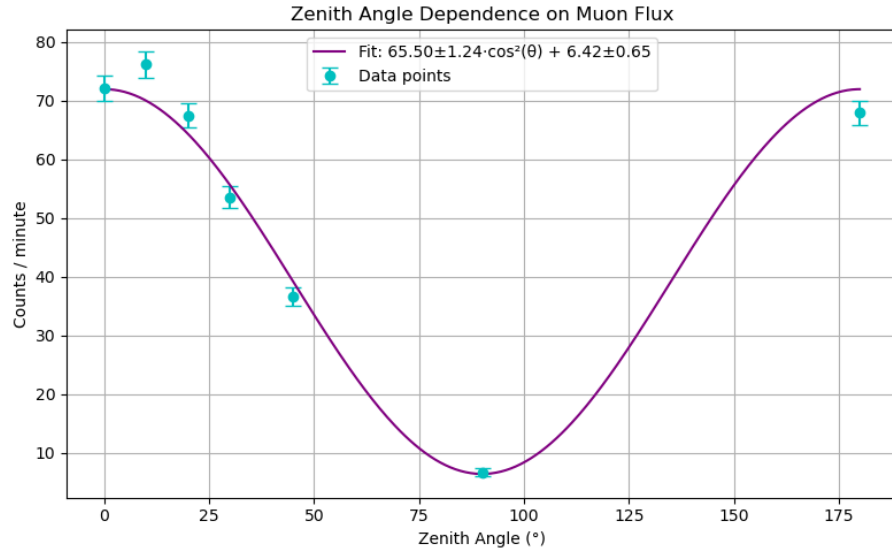


Figure 3.40: Dependence of muon flux on zenith angle. Data fitted with $f(\theta) = a \cos^2(\theta) + b$ as expected from the angular distribution of atmospheric muons.

The measured rates are summarized below:

Zenith Angle (°)	Counts/min	Error	Relative Trend
0	72.13	± 2.19	High
10	76.13	± 2.25	Highest
20	67.47	± 2.12	High
30	53.53	± 1.89	Medium
45	36.53	± 1.56	Low
90	6.67	± 0.67	Minimum
180	67.87	± 2.13	High

Table 3.9: Count rate of muons at different zenith angles. Data was integrated over a calibrated time window.

The data shows a strong dependence on angle, well-described by the fit:

$$f(\theta) = (65.50 \pm 1.24) \cdot \cos^2(\theta) + (6.42 \pm 0.65)$$

This follows the expected $\cos^2(\theta)$ distribution for muons at sea level, confirming the angular anisotropy of the muon flux.

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